# ל7 — Complex Amplitudes and the Quantum Sector of the Apex

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**Abstract**

**Situation.** ל0 named the apex generator $\mathcal{M}_x(\beta, c)$ of the structured categorical simplex as a Fourier-Mellin transform on $\mathbb{Z}/N$ of the centered log-ledger of a positive real ledger $x \in \mathbb{R}^N_{>0}$. ל1–ל6 worked the real-positive sector: the amplitude sphere $S^{N-1}_+ \subset \mathbb{R}^N_{>0}$, Born closure $p_i = x_i^2/\sum_j x_j^2$, the cumulant tower, the radical Helmert ladder. ל0 Proposition 3.7 established that on the real-positive sector the $n$-th-power Kummer cover is *trivial*, with the deck-group action collapsing because $\mu_n \cap \mathbb{R}_{>0} = \{1\}$. ל0 explicitly scoped the complex-amplitude extension — where the cover becomes non-trivial and the apex acquires phase structure — as a forward door (§6.1).

**Gap.** The complex-amplitude extension is not a cosmetic generalization. Replacing $\mathbb{R}^N_{>0}$ by $\mathbb{C}^N$ does four structurally distinct things at once: (i) it enlarges the closure symmetry from $S_N$ permutations to the unitary group $U(N)$; (ii) it activates the Kummer cover $\mu_n \subset S^1$ on each coordinate, giving a non-trivial deck group $(\mu_n)^N/\mu_n$ for the $n$-th-power closure; (iii) it introduces a phase fiber $T^{N-1} = U(1)^N/U(1)$ over each simplex point, with the real-positive sector recovered as the *decoherent* (phase-averaged) face; (iv) it forces the Fisher–Rao metric to be the real-positive restriction of a strictly larger projective metric, the Fubini–Study metric on $\mathbb{CP}^{N-1}$. The real-positive sector is one slice through a four-axis space (amplitude, phase, conductor, gauge). ל1–ל6 worked the slice; ל7 builds the ambient.

**Action.** We construct the complex apex on $\mathbb{C}^N$. The complex amplitude $z \in \mathbb{C}^N$ is positioned at three nested layers — the amplitude sphere $S^{2N-1} \subset \mathbb{C}^N$, the projective space $\mathbb{CP}^{N-1} = S^{2N-1}/U(1)$, and the probability simplex $\Delta^{N-1} = \mathbb{CP}^{N-1}/T^{N-1}$ — connected by the **Hopf** and **toric** fibrations. The Born closure

$$\pi_{\text{Born}} : \mathbb{C}^N \setminus \{0\} \longrightarrow \Delta^{N-1}, \qquad z \longmapsto \Big(\frac{|z_i|^2}{\sum_j |z_j|^2}\Big)_{i=1}^N$$

factors as $\pi_{\text{Born}} = \mu_{\text{toric}} \circ \text{Hopf} \circ \text{rad}$ through these layers, with $\mu_{\text{toric}}$ the moment map of the toric action of $T^{N-1}$ on $\mathbb{CP}^{N-1}$. We define the complex apex generator

$$\mathcal{M}_z(\beta, c) := \frac{1}{N} \sum_{s \in \mathbb{Z}/N} e^{\beta y_s(z)}\, \overline{\chi_c(s)}, \qquad y_s(z) := \log z_s - \tfrac{1}{N}\textstyle\sum_j \log z_j,$$

phrased precisely below as a holomorphic function of $\beta \in \mathbb{C}$ on each Riemann-surface branch of the complex log of $z$. We then prove the **decoherence theorem** identifying ל0's real-positive apex with the modulus-restriction of $\mathcal{M}_z$ from the complex amplitude to its modulus ledger, and the **Kummer-cover theorem** identifying the integer-$\beta$ samples on the complex side with sections of an explicit $\mu_N$-twisted line bundle over $\mathbb{CP}^{N-1}$.

**Result.** Six core claims of ל7:

1. **Three-layer factorization of Born closure.** The map $\pi_{\text{Born}} : \mathbb{C}^N \setminus \{0\} \to \Delta^{N-1}$ factors uniquely as $\mu_{\text{toric}} \circ \text{Hopf} \circ \text{rad}$ (applied right-to-left): radial scaling to the sphere $S^{2N-1}$, Hopf projection to $\mathbb{CP}^{N-1}$ (modding out global phase $U(1)$), then the toric moment map to the simplex (modding out relative phases $T^{N-1}$). Each layer has a well-defined symmetry quotient.
2. **Fubini–Study restricts to Fisher–Rao on the decoherent face (re-derivation).** The Fubini–Study metric $g_{\text{FS}}$ on $\mathbb{CP}^{N-1}$ restricts to one-quarter of the Fisher–Rao metric on the real-positive face under the square-root section. This is the classical FS / quantum-Fisher relation (Provost–Vallée 1980; Braunstein–Caves 1994; Bengtsson–Życzkowski 2017); we re-derive it in the categorical-simplex language to fix the attribution of the factor of 2 in ל1's $d_{FR} = 2\arccos\sum\sqrt{p_i q_i}$ to the **Born / square-root embedding** $p \mapsto \sqrt{p}$ (in agreement with ל6 §2), not the Hopf fibration: on the decoherent face the Hopf/global-phase quotient acts as an isometry, $d_{\text{FS}}([\sqrt{p}], [\sqrt{q}])$ equals the unit-sphere geodesic distance $\arccos\sum\sqrt{p_i q_i}$, and the doubling lives entirely in the square root, at all $N$.
3. **Complex apex generator.** Define $\mathcal{M}_z(\beta, c)$ on $\mathbb{C}^N$ via complex centered log-ledger $y_s = \log z_s - \frac{1}{N}\sum_j \log z_j$ (with a canonical branch on the positive sector and analytic continuation onto the cut

Riemann surface of the complex log). $\mathcal{M}_z$ is holomorphic in $\beta$ on each branch, and the real-positive evaluation recovers §0's $\mathcal{M}_x$.

4. **Decoherence theorem (modulus-phase reduction).** The complex apex generator splits termwise as
$$\mathcal{M}_z(\beta, c) = \frac{1}{N} \sum_s e^{\beta y_s^{(|z|)}} \cdot e^{i\beta y_s^{(\arg z)}} \overline{\chi_c(s)},$$
with $y_s^{(|z|)}$ the §0 centered log-ledger of the modulus ledger $|z| \in \mathbb{R}^N_{>0}$ and $y_s^{(\arg z)}$ the centered argument (Proposition 3.6). The **modulus-restriction** $z \mapsto |z|$ — the operational projection of complex amplitude to measurement-basis modulus — sets every phase factor to 1 and reduces $\mathcal{M}_z(\beta, c)$ to §0's $\mathcal{M}_{|z|}(\beta, c)$ exactly. The real-positive sector is *exactly* the locus $U(1) \cdot \mathbb{R}^N_{>0} \subset \Omega_N$ on which the phase deformation is trivial for generic $\beta$ (Proposition 3.6).
5. **Kummer-cover theorem.** The closure diagonal $\mathcal{M}_z(m, c)$ at integer $m$ has $(\mu_m)^N/\mu_m$ deck-group symmetry: replacing each $z_s$ by $\zeta_s z_s$ for $\zeta_s \in \mu_m$ leaves $|z_s|^{2m}$ invariant but multiplies the apex generator by $\prod_s \zeta_s^{\pm}$, encoding the Galois action on the moment diagonal as character-twisting on the cover.
6. **Qubit anchor at $N = 2$.** The qubit case $\mathbb{CP}^1 \cong S^2$ realizes the apex on the Bloch sphere, with the real-positive sector being the equatorial great circle $\{|0\rangle, |1\rangle\}$. The full Lee–Yang lattice of §0 Proposition 2.5 lifts to a phase-twisted lattice in the complex $\beta$-plane parameterized by the Bloch-vector orientation.

**Harvest.** The *phase structure* of pure-state finite-dimensional quantum mechanics on $\mathbb{C}^N$ is not an extra principle: it is the *cover* of the categorical-simplex apex — the same Fourier-Mellin transform, lifted from $\mathbb{R}^N_{>0}$ to $\mathbb{C}^N$ via the canonical Hopf-and-toric fibrations of complex projective space. We use the slogans *decoherence is closure of phase* and *quantization is the cover* in a deliberately scoped sense: the cover captures the pure-state phase redundancy — global $U(1)$, relative $T^{N-1}$, and the $\mu_N$ branch ambiguity of root closures — and **not** the non-commutativity of observables, measurement back-action, unitary dynamics, or tensor-product/entanglement structure, which sit in the forward doors of §7. Within that scope the real-positive sector is the decoherent face and the complex sector is the full bundle, and §6 Conjecture 4 ("complex amplitude extension") is now a structure theorem rather than a conjecture.

**Keywords:** finite-dimensional quantum mechanics, complex projective space, Fubini–Study metric, Hopf fibration, toric moment map, Born closure, Kummer cover, deck group, decoherence, phase fiber, qubit, $SU(2) \to SO(3)$, quantum Fisher information, Bures metric, finite-categorical Fourier-Mellin.

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*Decoherence is closure of phase. Quantization is the cover.*

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## Cross-references from prior papers

This paper builds on specific results from earlier papers of the ל-series; we list the dependencies explicitly.

| Cited result | Used for |
|---|---|
| ל0 §3.2 Definition 3.1 (additive apex generator $\mathcal{M}_x$) | Reduced from $\mathcal{M}_z$ in Proposition 3.5 (real-positive reduction) |
| ל0 §3.3 Theorem 3.5 (closure diagonal) | Underlies §5 Kummer cover theorem at integer $\beta = m$ |
| ל0 §3.4 Proposition 3.7 (real-sector Kummer triviality) | Special case of ל7 Proposition 5.6 |
| ל0 §2.6 Proposition 2.5 (Lee–Yang at $N = 2$) | Lifted to phase-twisted lattice in §6 Proposition 6.2 |
| ל1 (Fisher–Rao metric, $d_{FR} = 2 \arccos \sum \sqrt{p_i q_i}$) | Restriction of Fubini–Study in §2 Theorem 2.1 |
| ל6 (Born closure $p_i = x_i^2 / \sum x^2$) | Real-positive sector identified in §1.3 decoherent face |
| ל0 §3.3 (Bhattacharyya coefficient $\sum \sqrt{p_i q_i}$) | Theorem 2.1 proof |
| ל6 Conjecture 4 (complex amplitude extension) | Promoted to structure theorem by §5 and §4 |

ל7 introduces the complex amplitude apex $\mathcal{M}_z$, identified as the analytic continuation of ל0's $\mathcal{M}_x$ to the cut Riemann surface of complex log over $\Omega_N = (\mathbb{C}^\times)^N$.

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## Notation

We carry the notation of ל0 and add the complex-amplitude layer.

| Symbol | Meaning |
|---|---|
| $N$ | Number of categorical outcomes, integer $\geq 2$. |
| $z = (z_1, \dots, z_N)$ | Complex amplitude in $\mathbb{C}^N$. |
| $\lvert z \rvert^2$ | Componentwise modulus-squared: $\lvert z \rvert_i^2 = \lvert z_i \rvert^2 = z_i \bar{z}_i$. |

| Symbol | Meaning |
|---|---|
| $\|z\|$ | Euclidean norm: $\|z\| = (\sum_i \|z_i\|^2)^{1/2}$. |
| $\|z\|$ | Componentwise modulus ledger: $\|z\| = (\|z_1\|, \dots, \|z_N\|) \in \mathbb{R}^N_{>0}$. |
| $S^{2N-1}$ | Unit sphere in $\mathbb{C}^N$, $\{z : \|z\| = 1\}$. |
| $\mathbb{CP}^{N-1}$ | Complex projective space, $S^{2N-1}/U(1)$. |
| $\Delta^{N-1}$ | Probability simplex in $\mathbb{R}^N_{\geq 0}$ with $\sum_i p_i = 1$. |
| $\mathring{\Delta}^{N-1}$ | Open simplex (strictly positive interior). |
| $T^{N-1}$ | The relative-phase torus $U(1)^N/U(1)$. |
| $\Omega_N$ | Configuration space of nowhere-vanishing complex amplitudes, $(\mathbb{C}^\times)^N$. |
| $\pi_{\text{Born}}$ | Born closure $z \mapsto (\|z_i\|^2/\|z\|^2)_i : \mathbb{C}^N \setminus \{0\} \to \Delta^{N-1}$. |
| Hopf | Hopf projection $S^{2N-1} \to \mathbb{CP}^{N-1}$. |
| $\mu_{\text{toric}}$ | Toric moment map $\mathbb{CP}^{N-1} \to \Delta^{N-1}$. |
| $g_{\text{FS}}$ | Fubini–Study metric on $\mathbb{CP}^{N-1}$. |
| $g_{\text{FR}}$ | Fisher–Rao metric on $\mathring{\Delta}^{N-1}$. |
| $d_{\text{FS}}$, $d_{\text{FR}}$ | Fubini–Study, Fisher–Rao distance. |
| $\mathcal{M}_z(\beta, c)$ | Complex apex generator on $\Omega_N$ (Definition 3.4 below). |
| $\mathcal{M}_x(\beta, c)$ | ל0 real-positive apex generator on $\mathbb{R}^N_{>0}$. |
| $y_s(z)$ | Complex centered log-ledger: $y_s = \log z_s - \frac{1}{N}\sum_j \log z_j \in \mathbb{C}$. |
| $y_s^{(\|z\|)}$ | Real part: modulus log-ledger $\ln \|z_s\| - \frac{1}{N}\sum_j \ln \|z_j\| \in \mathbb{R}$. |
| $y_s^{(\arg z)}$ | Imaginary part: centered argument $\arg z_s - \frac{1}{N}\sum_j \arg z_j \in \mathbb{R}$. |
| $G(z)$ | Complex geometric mean, an $N$-th root of $\prod_j z_j$; multi-valued, single-valued on the Kummer cover. |
| $\widehat{z^m}_c$ | $c$-th Fourier coefficient of the componentwise $m$-th power: $\frac{1}{N}\sum_s z_s^m \overline{\chi_c(s)}$. |
| $\arg(z_s)$ | Principal argument of $z_s$, in $(-\pi, \pi]$. |
| $\mu_n$ | Group of $n$-th roots of unity in $\mathbb{C}^\times$. |
| $(\mu_n)^N/\mu_n$ | Componentwise deck group of the $n$-th-power Kummer cover. |
| $\widetilde{\Omega}_N$ | Degree-$N$ Kummer cover of $\Omega_N \cap \mathbb{CP}^{N-1}$ on which $G(z)$ is single-valued (Definition 5.1). |
| $\omega$ | Coordinate on the Kummer cover, $\omega^N = \prod_j z_j$. |
| $\rho_m$ | Deck-group character $\rho_m(\zeta) = \zeta^{-m}$ on $\mu_N$. |
| $\sigma_{\text{R+}}$ | Real-positive embedding $\mathbb{R}^N_{>0} \hookrightarrow \Omega_N$ (the no-phase section). |
| $\mathcal{R}$ | Modulus-restriction map $z \mapsto \|z\| : \Omega_N \to \mathbb{R}^N_{>0}$. |
| $\mathcal{P}$ | Computational-basis dephasing channel $\rho \mapsto \sum_s \langle s\|\rho\|s\rangle \|s\rangle\langle s\|$. |
| $L_\rho$ | Centered modular Hamiltonian: $\log \rho - \frac{1}{N}\mathrm{tr}(\log \rho)\, I_N$. |
| $\langle \psi \| \phi \rangle$ | Hermitian inner product on $\mathbb{C}^N$, $\sum_i \bar{\psi}_i \phi_i$. |
| $\|z\rangle$, $\langle z\|$ | Dirac notation for $z \in \mathbb{C}^N$ as ket / dual bra. |
| $\chi_c$ | Additive character on $\mathbb{Z}/N$: $\chi_c(s) = e^{2\pi i c s/N}$. |

| Symbol | Meaning |
|---|---|
| $\mathcal{O}(k)$ | Holomorphic line bundle on $\mathbb{CP}^{N-1}$ of degree $k$; $\mathcal{O}(-1)$ is the tautological bundle. |

**Acronyms.** Each is expanded on first use in the body; this table is for reference.

| Acronym | Expansion |
|---|---|
| AJLS | Ay–Jost–Lê–Schwachhöfer (modern functional-analytic synthesis of information geometry). |
| CRT | Chinese Remainder Theorem. |
| DFT | Discrete Fourier transform. |
| FR | Fisher–Rao (metric / distance on the simplex). |
| FS | Fubini–Study (metric / distance on $\mathbb{CP}^{N-1}$). |
| ODE | Ordinary differential equation. |
| QFI | Quantum Fisher information. |
| RMS | Root mean square ($r = 2$ power mean). |
| RH | Riemann hypothesis. |

**Conventions and dictionary.** Two notational conflicts deserve explicit attention.

(i) *Complex logarithm.* $\log z$ on $\mathbb{C} \setminus \{0\}$ is taken in the **principal branch** $\log z = \ln|z| + i \arg z$ with $\arg z \in (-\pi, \pi]$ and cut along the negative real axis, unless explicitly stated otherwise. Sheet shifts $\log z_s \mapsto \log z_s + 2\pi i k_s$ are tracked component-wise in §3.4.

(ii) *Gauge exponent $\beta$ across amplitude vs probability.* The exponent $\beta$ in ל7's $\mathcal{M}_z(\beta, c)$ refers to the **amplitude** ledger $z \in \mathbb{C}^N$, with closure $p_i = |z_i|^2/\|z\|^2$ producing probability under $\beta = 2$ on the modulus restriction. A density-matrix formulation acting on Born probabilities $p$ would use $\beta_\rho = \beta/2$ (§4.5); we write $\beta$ for the amplitude exponent throughout and explicitly note the half-factor where the probability convention appears.

(iii) *Source of the factor of* 2. The factor of 2 between Fisher–Rao and Fubini–Study distances in Theorem 2.1 is the **Born / square-root embedding** $p \mapsto \sqrt{p} : \Delta^{N-1} \to S^{N-1}_+$, not the Hopf fibration: on the decoherent face the Hopf/global-phase quotient acts as a local isometry (the vertical correction $|\langle \delta z | z \rangle|^2$ vanishes), so $d_{\mathrm{FS}}$ equals the unit-sphere geodesic distance and the doubling is the square root alone, at all $N$. This matches ל6 §2. The $SU(2) \to SO(3)$ double cover at $N = 2$ is the *global-phase* $z \sim -z$ quotient; it fixes the radius of the Fubini–Study Bloch sphere ($\mathbb{CP}^1 \cong S^2_{r=1/2}$) but does not supply the $d_{\mathrm{FR}}/d_{\mathrm{FS}}$ ratio (§2.2).

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## 1. The complex amplitude space and its three layers

### 1.1 From real-positive to complex

ל1–ל6 worked the real-positive sector: amplitudes $x \in \mathbb{R}^N_{>0}$, normalized to the positive face of the sphere $S^{N-1}_+ := S^{N-1} \cap \mathbb{R}^N_{\geq 0}$, with closure $p_i = x_i^2 / \sum_j x_j^2$ producing probability vectors in the simplex $\Delta^{N-1}$. The natural symmetry of the configuration is the permutation group $S_N$, acting by relabeling outcomes.

The complex amplitude extension replaces $\mathbb{R}^N$ by $\mathbb{C}^N$ and the positive face by the full unit sphere $S^{2N-1} \subset$

$\mathbb{C}^N$:

$$S^{2N-1} = \{z \in \mathbb{C}^N : \|z\|^2 = \sum_i |z_i|^2 = 1\}.$$

The symmetry enlarges to $U(N)$, the unitary group, which acts transitively on $S^{2N-1}$ and preserves Hermitian inner products $\langle z|w\rangle = \sum_i \bar{z}_i w_i$. The positive face $S^{N-1}_+$ sits inside $S^{2N-1}$ as the diagonal "real-positive corner" where all $z_i$ are real and positive.

The Born closure $z \mapsto (|z_i|^2/\|z\|^2)_i$ extends straight to $\mathbb{C}^N \setminus \{0\}$ (it depends only on moduli, so phases are quotiented out). The map factors through three nested quotients corresponding to three distinct symmetries, which we now describe.

### 1.2 The three layers

**Layer 1: radial scaling.** The cone $\mathbb{C}^N \setminus \{0\}$ deformation-retracts onto the sphere $S^{2N-1}$ via $z \mapsto z/\|z\|$. The symmetry quotiented out is the positive-real dilation $\mathbb{R}^\times_{>0}$, acting on $\mathbb{C}^N$ by $z \mapsto \lambda z$ for $\lambda > 0$. The Born closure is invariant under this action.

**Layer 2: global phase (Hopf fibration).** The unitary group $U(1) \subset \mathbb{C}^\times$ acts on $S^{2N-1}$ by uniform phase rotation: $z \mapsto e^{i\theta} z$. This is the **global phase symmetry** of quantum mechanics. The quotient is the complex projective space

$$\mathbb{CP}^{N-1} := S^{2N-1}/U(1),$$

and the projection $\mathrm{Hopf} : S^{2N-1} \to \mathbb{CP}^{N-1}$ is the **Hopf fibration** (in classical terminology, especially for $N = 2$ where $\mathrm{Hopf} : S^3 \to S^2$).

**Layer 3: relative phases (toric moment map).** The torus $T^N = U(1)^N$ acts on $\mathbb{C}^N$ componentwise: $(\theta_1, \dots, \theta_N) \cdot z = (e^{i\theta_1} z_1, \dots, e^{i\theta_N} z_N)$. The diagonal subgroup $U(1) \subset T^N$ acts by global phase (the symmetry of Layer 2). The quotient torus

$$T^{N-1} := T^N/U(1)$$

acts effectively on $\mathbb{CP}^{N-1}$ by relative phases, preserving moduli but rotating the relative arguments $\arg z_i - \arg z_1$. The quotient

$$\mu_{\mathrm{toric}} : \mathbb{CP}^{N-1} \to \Delta^{N-1}, \qquad [z] \mapsto \left(\frac{|z_i|^2}{\|z\|^2}\right)_i$$

is the **moment map** of this toric action in the standard sense of symplectic / toric geometry (Atiyah 1982; Guillemin–Sternberg 1982), and its image is the probability simplex.

**Theorem 1.1 (Three-layer factorization of Born closure).** The Born closure factors as a composition

$$\pi_{\mathrm{Born}} : \mathbb{C}^N \setminus \{0\} \xrightarrow{\mathrm{rad}} S^{2N-1} \xrightarrow{\mathrm{Hopf}} \mathbb{CP}^{N-1} \xrightarrow{\mu_{\mathrm{toric}}} \Delta^{N-1},$$

where each arrow is the quotient by a connected Lie subgroup of the structure group $\mathbb{R}^\times_{>0} \times U(1)^N$ of the action $\mathbb{C}^N \setminus \{0\} \circlearrowleft$ by positive scaling and componentwise phase: - rad quotients by $\mathbb{R}^\times_{>0}$ (radial scaling); - Hopf quotients by the diagonal $U(1)$ (global phase); - $\mu_{\mathrm{toric}}$ quotients by $T^{N-1} = U(1)^N/U(1)$ (relative phases).

The full structure group $\mathbb{R}^\times_{>0} \times U(1)^N$ acts on $\mathbb{C}^N \setminus \{0\}$ with quotient $\Delta^{N-1}$.

**Proof.** Each map is the quotient by its named group action; commutativity of these three (radial scaling, global phase, relative phases) makes the composition well-defined and unique. The structure group has dimension $1 + N = N + 1$, the fiber over a generic point of $\Delta^{N-1}$ (with no $p_i = 0$) has dimension $N + 1$, and the simplex has dimension $N - 1$; the total dimension $N + 1 + N - 1 = 2N = \dim_{\mathbb{R}} \mathbb{C}^N$ checks. □

### 1.3 The real-positive sector as decoherent face

The real-positive sector $\mathbb{R}^N_{>0} \subset \mathbb{C}^N$ corresponds to choosing all $\arg z_s = 0$, a single point in the relative-phase torus $T^{N-1}$ (the "all-zero phases" representative of each toric fiber). Equivalently, the positive face $S^{N-1}_+ \subset S^{2N-1}$ is a *section* of both the Hopf fibration (a choice of global-phase representative) and the toric fibration (a choice of relative-phase representative).

**Definition 1.2 (Decoherent face).** The **decoherent face** of the complex amplitude space is the real-positive sector $S^{N-1}_+ \subset S^{2N-1}$, equivalently the image of the section $p \mapsto (\sqrt{p_1}, \dots, \sqrt{p_N})$ of $\pi_{\text{Born}}$. The corresponding map $\Delta^{N-1} \to S^{N-1}_+$ is the **square-root section**.

The square-root section is the foundation of ל1's chart. ל7's framing places it as one *section* of the toric/Hopf fibrations among many: the all-phases-zero choice. Other sections exist (e.g., $z_s = \sqrt{p_s} e^{i\phi_s}$ for any phase pattern $\phi$), and the choice of section is a *gauge* in the precise sense of §1.4 below.

### 1.4 Phase as a gauge choice

The terminology of ל0 (the gauge corridor parameterized by $\beta$) and the terminology here ("global phase gauge" / "relative phase gauge") are not metaphorically related — they are instances of the same principle. The full structure group $\mathbb{R}^\times_{>0} \times U(1)^N$ acting on $\mathbb{C}^N \setminus \{0\}$ is, like the $L^r$ gauge corridor of ל0 §1.3, a choice of representative on a multi-parameter family of equivalence classes. ל0's gauge selection answered "what is additive?" by choosing $\beta$; ל7's gauge selection answers "what is the phase?" by choosing a section of the toric/Hopf fibrations. The two questions are independent (one is real, the other is complex), and the gauge corridor of ל0 lifts to a $(N+1)$-parameter gauge structure on $\mathbb{C}^N$ — one $\beta$ in $\mathbb{R}$ and $N$ phase angles in $T^N$, with the diagonal $U(1)$ being the global-phase quotient.

We do not develop the full $(N+1)$-parameter gauge corridor in ל7; we treat the phase axes as discrete (toric) symmetries with non-trivial $\mu_n$ subgroup structure at integer $\beta = m$ in the closure diagonal.

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## 2. The Fubini–Study metric and the projective angle

### 2.1 Fubini–Study and Fisher–Rao

The **Fubini–Study metric** $g_{\text{FS}}$ on $\mathbb{CP}^{N-1}$ is the unique (up to scale) $U(N)$-invariant Kähler metric, obtained by pushing forward the round metric on $S^{2N-1}$ through the Hopf fibration:

$$g_{\text{FS}}([z])(\delta_1[z], \delta_2[z]) = \text{Re}\Big( \frac{\langle \delta_1 z | \delta_2 z \rangle}{\|z\|^2} - \frac{\langle \delta_1 z | z \rangle \langle z | \delta_2 z \rangle}{\|z\|^4} \Big),$$

on tangent vectors orthogonal to $z$ in the Hermitian sense. The corresponding **Fubini–Study distance** is

$$d_{\text{FS}}([z], [w]) = \arccos\Big( \frac{|\langle z | w \rangle|}{\|z\| \|w\|} \Big).$$

The **Fisher–Rao metric** $g_{\text{FR}}$ on the open simplex $\mathring{\Delta}^{N-1}$ is the unique (up to scale) Riemannian metric invariant under stochastic transformations preserving relative entropy (Chentsov 1982; Amari–Nagaoka 2000), with Fisher–Rao distance

$$d_{\text{FR}}(p, q) = 2 \arccos \sum_i \sqrt{p_i q_i}.$$

**Theorem 2.1 (Fubini–Study restricts to Fisher–Rao on the decoherent face).** Let $\sigma : \mathring{\Delta}^{N-1} \to S^{N-1}_+$, $\sigma(p) = \sqrt{p} = (\sqrt{p_1}, \dots, \sqrt{p_N})$, be the square-root section. Then

$$d_{\text{FS}}([\sqrt{p}], [\sqrt{q}]) = \tfrac{1}{2} d_{\text{FR}}(p, q) \quad \text{for all } p, q \in \mathring{\Delta}^{N-1},$$

and the corresponding infinitesimal identity holds:

$$(\mathrm{Hopf} \circ \sigma)^* g_{\mathrm{FS}} = \tfrac{1}{4}\, g_{\mathrm{FR}} \quad \text{on } \mathring{\Delta}^{N-1}.$$

**Proof.** *(Distance.)* For $z = \sqrt{p}$ and $w = \sqrt{q}$ on the decoherent face, $\langle z|w\rangle = \sum_i \sqrt{p_i}\sqrt{q_i} \in \mathbb{R}_{\geq 0}$, so $|\langle z|w\rangle| = \sum_i \sqrt{p_i q_i}$ — the Bhattacharyya coefficient (Bhattacharyya 1943). Since $\|z\| = \|w\| = 1$ on the section,

$$d_{\mathrm{FS}}([\sqrt{p}], [\sqrt{q}]) = \arccos \sum_i \sqrt{p_i q_i} = \tfrac{1}{2} d_{\mathrm{FR}}(p, q),$$

using $d_{\mathrm{FR}}(p, q) = 2 \arccos \sum_i \sqrt{p_i q_i}$ (ל1).

*(Metric.)* Fix $p \in \mathring{\Delta}^{N-1}$ and a tangent vector $v \in T_p \Delta^{N-1} = \{v \in \mathbb{R}^N : \sum_i v_i = 0\}$. The Fisher–Rao metric is $g_{\mathrm{FR}}(p)(v, v) = \sum_i v_i^2 / p_i$ (Amari–Nagaoka 2000). The differential of the square-root section is $d\sigma(p)(v) = (v_1/(2\sqrt{p_1}), \ldots, v_N/(2\sqrt{p_N}))$, which is a real tangent vector to $S^{N-1}_+ \subset S^{2N-1} \subset \mathbb{C}^N$ at $z = \sqrt{p}$.

Pull back the Fubini–Study metric in its definition (§2.1) at $z = \sqrt{p}$ with $\delta z = d\sigma(p)(v)$. Since $\sqrt{p} \in \mathbb{R}^N$ and $\delta z \in \mathbb{R}^N$, the Hermitian inner product reduces to the real inner product:

$$\langle \delta z | \delta z \rangle = \sum_i \left( \frac{v_i}{2\sqrt{p_i}} \right)^2 = \frac{1}{4} \sum_i \frac{v_i^2}{p_i}, \qquad \langle \delta z | z \rangle = \sum_i \sqrt{p_i} \cdot \frac{v_i}{2\sqrt{p_i}} = \frac{1}{2} \sum_i v_i = 0$$

(the last equality from $\sum_i v_i = 0$, since $v$ is tangent to the simplex). Substituting into the Fubini–Study formula at $\|z\| = 1$,

$$g_{\mathrm{FS}}(\sqrt{p})(\delta z, \delta z) = \langle \delta z | \delta z \rangle - |\langle \delta z | z \rangle|^2 = \frac{1}{4} \sum_i \frac{v_i^2}{p_i} - 0 = \frac{1}{4} g_{\mathrm{FR}}(p)(v, v).$$

Polarizing the quadratic identity to a bilinear identity gives $(\mathrm{Hopf} \circ \sigma)^* g_{\mathrm{FS}} = \frac{1}{4} g_{\mathrm{FR}}$ as quadratic forms on $T_p \mathring{\Delta}^{N-1}$.

The metric identity $g_{\mathrm{FS}} = \frac{1}{4} g_{\mathrm{FR}}$ and the distance identity $d_{\mathrm{FS}} = \frac{1}{2} d_{\mathrm{FR}}$ are consistent under the integration of length: scaling the metric by $1/4$ scales lengths by $1/2$, and the geodesics on the decoherent face $S^{N-1}_+$ are great-circle arcs in $S^{2N-1}$ (the round-sphere geodesics restricted to the positive face), which are the square-root pullbacks of the Fisher–Rao geodesics on $\mathring{\Delta}^{N-1}$. □

The **factor of** 2 in ל1's Fisher–Rao distance is therefore the **Born / square-root embedding** $\sigma : p \mapsto \sqrt{p}$, *not* the Hopf fibration — exactly as ל6 §2 records ("the angle was waiting for the square"). The proof makes the separation explicit: the Hopf/global-phase vertical correction $-|\langle \delta z | z \rangle|^2$ vanishes identically on the decoherent face (since $\langle \delta z | z \rangle = \frac{1}{2} \sum_i v_i = 0$ for real $\delta z$ with $\sum_i v_i = 0$), so the Hopf submersion acts there as a *local isometry on the horizontal real section* — it contributes the identity, ratio 1. The full factor enters through $\langle \delta z | \delta z \rangle = \frac{1}{4} \sum_i v_i^2 / p_i$, the square-root differential $d\sigma(v)_i = v_i/(2\sqrt{p_i})$. Concretely: on the positive orthant $\sqrt{p}, \sqrt{q}$ are real and never antipodal (pairwise angle $\leq \pi/2$), so $d_{\mathrm{FS}}([\sqrt{p}], [\sqrt{q}]) = \arccos \sum_i \sqrt{p_i q_i}$ equals the unit-sphere geodesic distance, and the doubling sits entirely in the square-root map $\Delta^{N-1} \to S^{N-1}_+$, where it sends arc length on the probability simplex to twice the arc length on the amplitude sphere. The Hopf fibration *defines* the Fubini–Study metric; it does not supply the number.

**Status of Theorem 2.1 (re-derivation, not a new result).** The relation $g_{\mathrm{FS}} = \frac{1}{4} g_{\mathrm{FR}}$ on the decoherent face is classical — it is the standard correspondence between the Fubini–Study metric, the quantum Fisher information, and the classical Fisher–Rao metric on the diagonal/decoherent stratum (Provost–Vallée 1980; Braunstein–Caves 1994; Bengtsson–Życzkowski 2017). We include it as a clean re-derivation in the categorical-simplex language, for two purposes only: to fix the attribution of the factor of 2 to the Born square root (settling the apparent tension with the Hopf reading), and to set up the genuinely new object of this paper — the Kummer-cover reading of the closure diagonal in §5, where the integer-$\beta$ apex generators become sections of an explicit $\mu_N$-twisted line bundle. §5, not Theorem 2.1, carries the novel content of ל7's metric/geometry layer.

### 2.2 The qubit case $N = 2$

For $N = 2$, $\mathbb{CP}^1 \cong S^2$ (the Bloch sphere), and the Hopf fibration is the classical $S^3 \to S^2$ with fiber $S^1$. The decoherent face $S^1_+ \subset S^3$ projects through Hopf to a great circle on $S^2$ (the "equator" connecting $|0\rangle$ and $|1\rangle$). It is worth separating two distinct factors of 2 that both appear here, since they are easily conflated:

- The $d_{\mathrm{FR}} = 2d_{\mathrm{FS}}$ **doubling** is the Born / square root, as in Theorem 2.1. Even at $N = 2$ it is the squaring: a real amplitude $\cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$ sits at Hilbert half-angle $\theta/2$, while its Born image sits at Bloch polar angle $\theta$, with $\cos\theta = p_0 - p_1$. This factor persists at all $N$.
- The $SU(2) \to SO(3)$ **double cover** is the *global-phase* quotient $z \sim -z$ (the $\theta = \pi$ fiber element). It controls the **radius** of the Fubini–Study Bloch sphere — $\mathbb{CP}^1$ with $g_{\mathrm{FS}}$ is $S^2$ of radius $\frac{1}{2}$, not the unit sphere — and is a low-dimensional coincidence of $N = 2$ that does not extend.

The $d_{\mathrm{FR}}/d_{\mathrm{FS}}$ ratio is the first item, not the second: on the decoherent face the Hopf/global-phase quotient acts as an isometry (Theorem 2.1 proof), so the $SU(2) \to SO(3)$ identification does not supply the doubling.

We return to the qubit anchor in §6.

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## 3. The complex apex generator $\mathcal{M}_z(\beta, c)$

This section is the technical centerpiece. We extend Ɔ0's apex generator from $\mathbb{R}^N_{>0}$ to $\mathbb{C}^N \setminus \bigcup_s \{z_s = 0\}$, define the complex centered log-ledger by componentwise complex logarithm, identify the resulting $\mathcal{M}_z(\beta, c)$ as a holomorphic function of $\beta \in \mathbb{C}$ on each chosen branch, derive its reduction to Ɔ0's $\mathcal{M}_x$ on the decoherent face, and characterize the branch structure (which becomes the literal Kummer-cover data of §5).

### 3.1 The complex centered log-ledger

We work on the open subset

$$\Omega_N := \{z \in \mathbb{C}^N : z_s \neq 0 \text{ for every } s\} = (\mathbb{C}^\times)^N \subset \mathbb{C}^N,$$

the configuration space of nowhere-vanishing complex amplitudes. The closure of $\Omega_N$ inside $\mathbb{C}^N$ adds the coordinate hyperplanes $\{z_s = 0\}$, on which one or more amplitudes vanish; these correspond to *boundary* simplex points (some $p_s = 0$ under Born closure) and require separate treatment.

**Branch choice.** For each $s$, choose a branch cut of the complex logarithm $\log z_s$. The default choice is the **principal branch**, $\arg z_s \in (-\pi, \pi]$ with the cut along the negative real axis. On the real-positive sector $z_s > 0$ this branch is real-analytic and agrees with $\ln z_s$. We will track sheet shifts $\log z_s \mapsto \log z_s + 2\pi i k_s$ ($k_s \in \mathbb{Z}$) explicitly in §3.5.

**Definition 3.1 (Complex centered log-ledger).** For $z \in \Omega_N$ with a chosen branch of $\log z_s$ for each coordinate $s$,

$$y_s(z) := \log z_s - \frac{1}{N}\sum_{j=1}^{N} \log z_j, \qquad s = 1, \dots, N.$$

The vector $y = (y_1, \dots, y_N) \in \mathbb{C}^N$ satisfies $\sum_s y_s = 0$, hence lies in the complex mean-zero hyperplane $H_N \otimes \mathbb{C}$.

**Lemma 3.2 (Gauge invariance and modulus-phase splitting).**

(a) *Positive-scaling invariance.* For $\lambda \in \mathbb{R}_{>0}$, $y_s(\lambda z) = y_s(z)$. So $y$ depends only on the ray $[z] \in \mathbb{C}^N/\mathbb{R}^\times_{>0}$.

(b) *Global-phase invariance.* For $\alpha \in \mathbb{R}$, $y_s(e^{i\alpha} z) = y_s(z)$. So $y$ descends to the global-phase quotient $\mathbb{C}^N \setminus \{0\}/(\mathbb{R}^\times_{>0} \times U(1))$ (modulo branch choices, addressed in §3.4).

(c) *Modulus-phase splitting.* Writing $z_s = |z_s| e^{i \arg z_s}$ on the principal branch,

$$y_s(z) = y_s^{(|z|)} + i\, y_s^{(\arg z)}, \qquad \text{where}$$
$$y_s^{(|z|)} := \ln |z_s| - \tfrac{1}{N} \sum_j \ln |z_j| \in \mathbb{R},$$
$$y_s^{(\arg z)} := \arg z_s - \tfrac{1}{N} \sum_j \arg z_j \in \mathbb{R}.$$

The real part is the Ł0 centered log-ledger of the modulus ledger $|z| = (|z_1|, \dots, |z_N|) \in \mathbb{R}^N_{>0}$. The imaginary part is the **centered argument**, the centered version of the phase pattern.

**Proof.** (a) $\log(\lambda z_s) = \log \lambda + \log z_s$ on the principal branch, and the constant $\log \lambda$ cancels under centering. (b) $\log(e^{i\alpha} z_s) = i\alpha + \log z_s$ similarly cancels. (c) $\log z_s = \ln |z_s| + i \arg z_s$ on the principal branch; linearity of the centering operation separates real and imaginary parts. □

### 3.2 Definition of the complex apex generator

**Definition 3.3 (Complex apex generator).** For $z \in \Omega_N$ with a chosen branch of $\log z_s$ for each $s$, $\beta \in \mathbb{C}$, and $c \in \widehat{\mathbb{Z}/N}$, the **complex apex generator** is

$$\boxed{\mathcal{M}_z(\beta, c) := \frac{1}{N} \sum_{s \in \mathbb{Z}/N} e^{\beta y_s(z)}\, \overline{\chi_c(s)},}$$

where $y_s(z)$ is the complex centered log-ledger of Definition 3.1.

The placement of $\beta$ in the exponent and $c$ in the character index is identical to Ł0 Definition 3.1; the only change is that $y_s$ is now complex-valued.

**Proposition 3.4 (Holomorphy in $\beta$).** For each fixed branch choice and each fixed $z \in \Omega_N$, $\mathcal{M}_z(\cdot, c) : \mathbb{C} \to \mathbb{C}$ is entire in $\beta \in \mathbb{C}$.

**Proof.** Each summand $e^{\beta y_s(z)} \overline{\chi_c(s)}$ is entire in $\beta$ (composition of the linear map $\beta \mapsto \beta y_s(z)$ with the entire exponential, multiplied by a $\beta$-independent constant). A finite sum of entire functions is entire. □

**Reduction to Ł0 on the real-positive sector.**

**Proposition 3.5 (Real-positive reduction).** If $z \in \mathbb{R}^N_{>0} \subset \Omega_N$ and the principal branch is chosen for every coordinate, then $y_s(z) \in \mathbb{R}$ for every $s$, and

$$\mathcal{M}_z(\beta, c) = \mathcal{M}_x(\beta, c) \quad \text{with } x := z \in \mathbb{R}^N_{>0},$$

where $\mathcal{M}_x$ is Ł0's real-positive apex generator. In particular, the entire holomorphic family $\{\mathcal{M}_z(\cdot, c) : z \in \Omega_N\}$ extends Ł0's $\{\mathcal{M}_x(\cdot, c) : x \in \mathbb{R}^N_{>0}\}$ off the real-positive sector.

**Proof.** For $z_s > 0$, $\log z_s = \ln z_s \in \mathbb{R}$ on the principal branch, so $y_s(z) \in \mathbb{R}$. Substituting into Definition 3.3 and comparing to Ł0 Definition 3.1 yields the equality. The holomorphic extension follows from Proposition 3.4 plus the fact that $\Omega_N$ is an open complex manifold containing $\mathbb{R}^N_{>0}$ as a real-analytic submanifold. □

### 3.3 Modulus-phase factorization

Combining Lemma 3.2(c) with Definition 3.3 yields a clean modulus-phase factorization of the complex apex.

**Proposition 3.6 (Modulus-phase factorization).** For $z \in \Omega_N$,

$$\mathcal{M}_z(\beta, c) = \frac{1}{N} \sum_s e^{\beta y_s^{(|z|)}}\, e^{i\beta y_s^{(\arg z)}}\, \overline{\chi_c(s)}.$$

The first exponential factor depends only on the moduli $|z| = (|z_1|, \dots, |z_N|)$ and reproduces §0's $\mathcal{M}_x$ structure (at $x = |z|$); the second exponential factor depends only on the centered argument $y_s^{(\arg z)}$ and encodes the phase content. In particular, for every $\beta \in \mathbb{C} \setminus (2\pi\mathbb{Q})$,

$$\big[\, \mathcal{M}_z(\beta, c) = \mathcal{M}_{|z|}(\beta, c) \ \text{ for all } c \in \widehat{\mathbb{Z}/N} \,\big] \quad \Longleftrightarrow \quad z \in U(1) \cdot \mathbb{R}^N_{>0}$$

(a global-phase rotation of the real-positive sector). The quantifier "for all $c$" is essential for ($\Rightarrow$): at a single fixed $c$ the character sum can vanish by accidental cancellation without the phases being trivial.

**Proof.** Direct from Definition 3.1 and Lemma 3.2(c): $y_s = y_s^{(|z|)} + i y_s^{(\arg z)}$, so $e^{\beta y_s} = e^{\beta y_s^{(|z|)}} e^{i\beta y_s^{(\arg z)}}$. Substituting into Definition 3.3 yields the factorization. For ($\Leftarrow$), $z \in U(1) \cdot \mathbb{R}^N_{>0}$ gives $y_s^{(\arg z)} = 0$ for every $s$, so every phase factor is 1 and $\mathcal{M}_z(\beta, c) = \mathcal{M}_{|z|}(\beta, c)$ for all $c$. For ($\Rightarrow$), equality at *all* $c$ means the two functions $s \mapsto e^{\beta y_s}$ and $s \mapsto e^{\beta y_s^{(|z|)}}$ on $\mathbb{Z}/N$ have identical Fourier coefficients, hence are equal pointwise: $e^{\beta y_s^{(|z|)}} e^{i\beta y_s^{(\arg z)}} = e^{\beta y_s^{(|z|)}}$ for every $s$. Since $e^{\beta y_s^{(|z|)}} \neq 0$, every phase factor $e^{i\beta y_s^{(\arg z)}} = 1$, i.e., $\beta y_s^{(\arg z)} \in 2\pi\mathbb{Z}$ for every $s$. For $\beta \notin 2\pi\mathbb{Q}$, this forces $y_s^{(\arg z)} = 0$, equivalently $z$ has uniform phase, equivalently $z \in U(1) \cdot \mathbb{R}^N_{>0}$. (For $\beta \in 2\pi\mathbb{Q}$ the lattice of "accidental zeros" $y^{(\arg z)} \in (2\pi/\beta)\mathbb{Z}^N$ admits additional solutions; these are precisely the $\mu_N$-translates that the Kummer-cover formalism of §5 organizes.) □

**Remark 3.7 (Comparison to the real-positive apex).** Proposition 3.6 says that $\mathcal{M}_z(\beta, c)$ splits as a *deformation* of $\mathcal{M}_{|z|}(\beta, c)$ by phase factors $e^{i\beta y_s^{(\arg z)}}$. The real-positive sector is *exactly* the fixed locus of "no deformation" for generic $\beta$, equal to $U(1) \cdot \mathbb{R}^N_{>0}$ on the global-phase quotient. This is the cleanest formulation of the slogan **"the real-positive sector is the no-phase section"** of the complex apex.

### 3.4 Branch structure: the Kummer cover seen from $\beta$

Definition 3.1 depends on a branch choice for each $\log z_s$. We characterize how $\mathcal{M}_z(\beta, c)$ changes when the branch shifts.

**Proposition 3.8 (Branch-shift formula).** Replace the chosen branch of $\log z_s$ by $\log z_s + 2\pi i k_s$ for some integer shift vector $k = (k_1, \dots, k_N) \in \mathbb{Z}^N$. The centered log-ledger transforms as

$$y_s(z) \longmapsto y_s(z) + 2\pi i (k_s - \bar{k}), \qquad \bar{k} := \tfrac{1}{N} \textstyle\sum_j k_j,$$

and the apex generator transforms as

$$\mathcal{M}_z(\beta, c) \longmapsto \frac{1}{N} \sum_s e^{2\pi i \beta (k_s - \bar{k})}\, e^{\beta y_s(z)}\, \overline{\chi_c(s)}.$$

In particular:

(a) If $k = (k_0, k_0, \dots, k_0)$ is the uniform shift, $k_s - \bar{k} = 0$ for every $s$ and $\mathcal{M}_z$ is invariant. (Uniform branch shift is the global-phase ambiguity, already quotiented in Lemma 3.2(b).)

(b) For non-uniform $k$ and $\beta \in \mathbb{C}$, the apex generator is multi-valued on the global-phase quotient $\mathbb{C}^N \setminus \{0\}/U(1)$ unless $\beta(k_s - \bar{k}) \in \mathbb{Z}$ for every admissible non-uniform $k$.

(c) For integer $\beta = m \in \mathbb{Z}$, the condition in (b) reduces to $m(k_s - \bar{k}) \in \mathbb{Z}$ for every $k \in \mathbb{Z}^N$. Taking $k = e_s$ (the $s$-th standard basis vector) gives $m(1 - 1/N) \in \mathbb{Z}$, i.e., $N \mid m$. Hence $\mathcal{M}_z(m, c)$ is single-valued on the global-phase quotient (and thus on $\mathbb{CP}^{N-1}$ after the toric quotient) precisely when $N \mid m$; for $N \nmid m$ it is multi-valued, with deck group the cyclic factor $\mu_N$ at each branch shift.

**Proof.** The branch-shift transformation rule is immediate from $\log z_s \mapsto \log z_s + 2\pi i k_s$ and the linearity of the centering operation. Substituting into Definition 3.3 produces the stated form. (a) is immediate. (b) follows because $\mathcal{M}_z$ as a function on $\mathbb{C}^N$ (with chosen branches) descends to the global-phase quotient iff branch shifts preserve its value. (c) reduces the integrality condition to a single test on a basis vector. □

**Remark 3.9 (Cover structure preview).** Proposition 3.8 is the precise statement of ל0 §3.4's "naturally organized by the Kummer cover" in the complex case. The closure diagonal at integer $\beta = m$ acquires the deck group $\mu_N/\mu_{\gcd(N,m)}$ at each coordinate (the obstruction to $m(k_s - \bar{k}) \in \mathbb{Z}$ for arbitrary $k_s$), and the global deck group of the cover $\widetilde{\mathbb{CP}^{N-1}} \to \mathbb{CP}^{N-1}$ on which $\mathcal{M}_z(m, \cdot)$ is single-valued is $(\mu_N)^N/\mu_N$ modulo the integrality condition $\sum_s k_s \equiv 0 \pmod N$. We develop the full deck group action in §5 (the **Kummer-cover theorem**); the operational message of §3 is that the cover structure is *already visible* in the branch-shift behavior of $\mathcal{M}_z$, and reduces to the trivial cover precisely on the real-positive sector where $\arg z_s = 0$ uniquely fixes every branch.

### 3.5 Summary: the complex apex as analytic continuation of the real apex

The complex apex generator $\mathcal{M}_z(\beta, c)$ is the analytic continuation of ל0's $\mathcal{M}_x(\beta, c)$ from $\mathbb{R}^N_{>0}$ to $\Omega_N = (\mathbb{C}^\times)^N$, treated as a function on the universal cover of $\Omega_N$ in $z$ (via branch choices for $\log z_s$) and entire in $\beta$ on each fixed branch. Three structural features distinguish it from $\mathcal{M}_x$:

1. **Phase factor.** Proposition 3.6 gives the modulus-phase splitting; the complex apex deforms the real apex by phase exponentials.
2. **Branch dependence.** Proposition 3.8 gives the branch-shift formula; the deck group of the Kummer cover acts on the integer-$\beta$ samples.
3. **Holomorphy in $\beta$.** Proposition 3.4 lifts ל0 Theorem 2.2's real-analytic statement to genuine holomorphy on all of $\mathbb{C}$.

The decoherence theorem of §4 connects feature (1) to ל0's $\mathcal{M}_x$ via phase-modulus separation; the Kummer-cover theorem of §5 makes feature (2) into a literal Galois statement on the integer samples; and the Lee–Yang lattice of ל0 Proposition 2.5 lifts via feature (3) to a phase-twisted lattice on the qubit case (§6).

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## 4. The decoherence theorem

This section is the bridge between ל7's complex apex generator $\mathcal{M}_z$ and ל0's real-positive apex $\mathcal{M}_x$. The bridge is the **modulus-restriction map** $\mathcal{R} : z \mapsto |z|$, the operational measurement-basis projection that ignores phase content and keeps only the Born probabilities. Theorem 4.2 below states that $\mathcal{R}$ pulls the complex apex back to the classical apex on the modulus ledger; Proposition 4.3 quantifies the gap between the two as a phase-deformation factor; §4.4 explains why the naive $T^{N-1}$-average of $\mathcal{M}_z$ is the *wrong* candidate for decoherence at integer $\beta$; §4.5 promotes the pure-state case to a density-matrix decoherence theorem (Theorem 4.5) and scopes the mixed-state generalization as a forward door.

### 4.1 The modulus-restriction map

**Definition 4.1 (Modulus-restriction).** The **modulus-restriction map** on the complex amplitude configuration space is

$$\mathcal{R} : \Omega_N \longrightarrow \mathbb{R}^N_{>0}, \qquad \mathcal{R}(z) := (|z_1|, |z_2|, \dots, |z_N|).$$

The map descends to the global-phase quotient $\Omega_N/U(1) \subset \mathbb{CP}^{N-1}$ (since $|z_s|$ is invariant under $z \mapsto e^{i\alpha}z$) and further to the toric quotient $\Omega_N/(U(1) \times T^{N-1})$, the open simplex $\mathring{\Delta}^{N-1}$, where it reduces to $z \mapsto (|z_i|^2/\|z\|^2)_i = \pi_{\text{Born}}(z)$ followed by the square-root section $p \mapsto \sqrt{p}$.

Operationally, $\mathcal{R}$ is the **measurement-basis modulus projection**: given a complex amplitude $z \in \mathbb{C}^N$, return the vector of moduli $|z| \in \mathbb{R}^N_{>0}$. This is the natural projection associated to a von Neumann measurement in the computational basis, in which the measurement outcomes are the moduli (or, after Born closure, the probabilities), and the phases are erased.

### 4.2 The decoherence theorem

**Theorem 4.2 (Decoherence as modulus-restriction).** For every $z \in \Omega_N$, every $\beta \in \mathbb{C}$, and every character $c \in \widehat{\mathbb{Z}/N}$, the apex generator of the dephased (no-phase) ledger $\sigma_{\mathrm{R}+}(\mathcal{R}(z))$ equals the classical apex of the modulus ledger:

$$\boxed{\mathcal{M}_{\sigma_{\mathrm{R}+}(\mathcal{R}(z))}(\beta, c) = \mathcal{M}_{|z|}(\beta, c),}$$

where $\mathcal{R}(z) = |z|$ is the modulus-restriction (Definition 4.1), $\sigma_{\mathrm{R}+} : \mathbb{R}^N_{>0} \hookrightarrow \Omega_N$ is the real-positive embedding (the canonical "no-phase" section), and $\mathcal{M}_{|z|}$ is Ϟ0's real-positive apex generator applied to the modulus ledger $|z| \in \mathbb{R}^N_{>0}$. Equivalently, the diagram

$$\begin{array}{ccc} \Omega_N & \xrightarrow{\mathcal{M}_{(-)}(\beta,c)} & \mathbb{C} \\ {\scriptstyle \sigma_{\mathrm{R}+}\circ\mathcal{R}}\downarrow & & \| \\ \mathbb{R}^N_{>0} & \xrightarrow{\mathcal{M}_{(-)}(\beta,c)} & \mathbb{C} \end{array}$$

commutes: restricting to moduli and re-embedding as a zero-phase amplitude leaves the apex generator unchanged.

**Proof.** Modulus-restriction followed by real-positive embedding sends $z \in \Omega_N$ to $\sigma_{\mathrm{R}+}(\mathcal{R}(z)) = |z| \in \mathbb{R}^N_{>0}$, viewed as a complex amplitude with zero phase. By Lemma 3.2(c), the complex centered log-ledger of $|z|$ has $y_s^{(\arg|z|)} = 0$ for every $s$, so $y_s(|z|) = y_s^{(|z|)} \in \mathbb{R}$. By Proposition 3.5, $\mathcal{M}_{|z|}(\beta, c) = \mathcal{M}_x(\beta, c)$ with $x = |z| \in \mathbb{R}^N_{>0}$. Hence $\mathcal{M}_{\sigma_{\mathrm{R}+}(\mathcal{R}(z))}(\beta, c) = \mathcal{M}_{|z|}(\beta, c)$. □

Theorem 4.2 is the precise sense in which **decoherence is the modulus-restriction**: the operation $\mathcal{R}\circ\sigma_{\mathrm{R}+}$ (restrict to moduli, then re-embed as zero-phase amplitudes) erases the phase content of $z$ and collapses the complex apex to the classical apex. Every integer-$\beta$ closure sample (Ϟ0 Theorem 3.5) and every Taylor-$\beta$ moment sample (Ϟ0 Theorem 3.3) of the complex apex is preserved under this operation — the operation kills *only* the phase deformation.

### 4.3 The phase deformation as the coherence content

**Proposition 4.3 (Quantitative decoherence gap).** For $z \in \Omega_N$, the difference between the complex apex and its decoherent restriction is

$$\mathcal{M}_z(\beta, c) - \mathcal{M}_{|z|}(\beta, c) = \frac{1}{N} \sum_{s \in \mathbb{Z}/N} e^{\beta y_s^{(|z|)}} \left(e^{i\beta y_s^{(\arg z)}} - 1\right) \overline{\chi_c(s)},$$

where each $(e^{i\beta y_s^{(\arg z)}} - 1)$ term encodes the deviation of the $s$-th coordinate's phase from the real-positive section.

In particular:

(a) *Vanishing on the real-positive sector.* If $z \in U(1) \cdot \mathbb{R}^N_{>0}$ (a global-phase rotation of the real-positive sector), every $y_s^{(\arg z)} = 0$ (by Lemma 3.2(c) plus global-phase invariance Lemma 3.2(b)) and the gap vanishes.

(b) *Linear regime.* For small $|\beta y_s^{(\arg z)}| \ll 1$,

$$\mathcal{M}_z(\beta, c) - \mathcal{M}_{|z|}(\beta, c) = i\beta \cdot \frac{1}{N} \sum_s e^{\beta y_s^{(|z|)}} y_s^{(\arg z)} \overline{\chi_c(s)} + O(\beta^2 |y^{(\arg z)}|^2),$$

the leading-order **coherence content** of the complex apex — a character-twisted moment of the centered argument $y^{(\arg z)}$, modulated by the classical apex weights $e^{\beta y_s^{(|z|)}}$.

**Proof.** Subtracting the modulus-restricted apex from the modulus-phase factorization of Proposition 3.6,

$$\mathcal{M}_z(\beta, c) - \mathcal{M}_{|z|}(\beta, c) = \frac{1}{N} \sum_s e^{\beta y_s^{(|z|)}} (e^{i\beta y_s^{(\arg z)}} - 1) \overline{\chi_c(s)},$$

which is the stated identity. (a) Each phase factor vanishes when $y_s^{(\arg z)} = 0$. (b) First-order Taylor: $e^{i\beta y_s^{(\arg z)}} - 1 = i\beta y_s^{(\arg z)} + O(\beta^2 |y^{(\arg z)}|^2)$. □

The coherence content $\mathcal{M}_z - \mathcal{M}_{|z|}$ measures the "non-classical" portion of the complex apex generator. In the quantum-information reading, it is the **interference content** of the state $|z\rangle$ relative to the computational basis $\{|s\rangle\}_{s=0}^{N-1}$: a state with zero coherence content has only diagonal (Born) statistics, while a state with non-zero coherence content has off-diagonal phase-sensitive structure that survives the apex measurement.

### 4.4 Why naive phase-averaging is the wrong decoherence operation

It is tempting to define decoherence as uniform averaging over the phase fiber $T^{N-1}$ — the symmetry-group quotient associated to "ignoring phases." We record explicitly that this is *not* an operation that recovers the classical apex, and that the modulus-restriction of Theorem 4.2 (which does recover ל0's apex on the modulus ledger) is the correct operational analog of measurement-basis decoherence.

**Proposition 4.4 (Phase-fiber averaging kills integer modes).** Fix a simplex point $p \in \mathring{\Delta}^{N-1}$ and parameterize the unit-sphere phase fiber over $p$ by $z_s = \sqrt{p_s}\, e^{i\phi_s}$ with $\phi = (\phi_1, \ldots, \phi_N) \in T^N$ modded by global phase to give $\phi \in T^{N-1}$. For every $\beta = m \in \mathbb{Z} \setminus \{0\}$ and every $c \in \widehat{\mathbb{Z}/N}$,

$$\frac{1}{(2\pi)^{N-1}} \int_{T^{N-1}} \mathcal{M}_z(m, c)\, d\phi = 0.$$

**Proof.** By the modulus-phase factorization (Proposition 3.6),

$$\mathcal{M}_z(m, c) = \frac{1}{N} \sum_s e^{m y_s^{(|z|)}}\, e^{im y_s^{(\arg z)}}\, \overline{\chi_c(s)},$$

where $y_s^{(|z|)}$ depends only on $p$ and is independent of $\phi$, while $y_s^{(\arg z)} = \phi_s - \frac{1}{N}\sum_j \phi_j$ depends linearly on $\phi$. Integrating term-by-term:

$$\int_{T^{N-1}} e^{im y_s^{(\arg z)}} \frac{d\phi}{(2\pi)^{N-1}} = \int_{T^{N-1}} e^{im(\phi_s - \bar{\phi})} \frac{d\phi}{(2\pi)^{N-1}}.$$

Parameterize $T^{N-1}$ by $(\phi_1, \ldots, \phi_{N-1}) \in T^{N-1}$ with $\phi_N = -\sum_{j<N} \phi_j$ (gauge-fixing global phase by $\sum_s \phi_s = 0$, so $\bar{\phi} = 0$). Then $e^{im(\phi_s - \bar{\phi})} = e^{im\phi_s}$, and for $1 \le s < N$,

$$\int_{T^{N-1}} e^{im\phi_s} \frac{d\phi}{(2\pi)^{N-1}} = \frac{1}{2\pi} \int_0^{2\pi} e^{im\phi_s} d\phi_s \cdot \prod_{j \neq s, N} \frac{1}{2\pi} \int_0^{2\pi} d\phi_j = \delta_{m,0} \cdot 1 = 0$$

for $m \neq 0$ (the $\phi_s$-integral is zero for any integer $m \neq 0$). For $s = N$: $\phi_N = -\sum_{j<N} \phi_j$, so $e^{im\phi_N} = \prod_{j<N} e^{-im\phi_j}$, and each $\phi_j$-integral is again zero for $m \neq 0$. Hence every summand vanishes after integration. □

Proposition 4.4 has a clean operational interpretation: **uniform averaging over the relative-phase torus annihilates every integer-$\beta$ mode of $\mathcal{M}_z$**, including modes (like $\beta = 2$, the Born/RMS rung of ל6) at which ל0's $\mathcal{M}_x$ is generically non-zero. Hence phase-fiber averaging cannot reproduce the classical apex; it produces zero (at non-zero integer $\beta$). The classical-apex content is *not* the phase-average of the complex apex — it is the phase-zero *section* of the complex apex, which is what modulus-restriction extracts.

The structural distinction:

- **Phase-fiber averaging** integrates $\mathcal{M}_z$ democratically over $T^{N-1}$, treating every phase configuration as equally weighted. This is a *measure-theoretic* operation on the complex amplitude space, killing all integer-Fourier-mode content of the apex.
- **Modulus-restriction** picks the unique zero-phase representative of each modulus orbit, treating the real-positive sector as a privileged *section* (not a measure-theoretic average). This is a *geometric* operation, preserving the full integer-$\beta$ content of ϟ0's apex.

These two operations are not the same and have opposite content: the first erases everything classical, the second isolates everything classical. Decoherence in the operational sense of measurement-basis modulus projection is the second, not the first.

**Open Problem 4.4 (Uniqueness of modulus-restriction as classical retract).** Characterize the family of "phase-erasing" maps $\Phi : \Omega_N \to \Omega_N$ — natural, $U(N)$-equivariant up to a chosen basis, idempotent — for which $\mathcal{M}_{\Phi(z)}(\beta, c) = \mathcal{M}_{|z|}(\beta, c)$ for all $(\beta, c)$. We expect modulus-restriction $\sigma_{\mathrm{R}+} \circ \mathcal{R}$ to be the unique such retract in the operationally meaningful sense (preserving moduli, killing only phases), but a precise functorial characterization is open. See §7 for the connection to the dephasing channel on density matrices.

### 4.5 Density-matrix decoherence on the pure-state stratum

The modulus-restriction formulation of Theorem 4.2 lives at the level of pure-state amplitudes $z \in \Omega_N$. We promote it to a density-matrix statement on the **pure-state stratum** and explicitly scope the mixed-state generalization as a forward door.

The operational decoherence on density matrices is the computational-basis **dephasing channel**

$$\mathcal{P} : \rho \longmapsto \sum_{s=0}^{N-1} \langle s|\rho|s\rangle \, |s\rangle\langle s|,$$

which sets every off-diagonal entry $\rho_{ss'}$ $(s \neq s')$ to zero while preserving the diagonal (the Born probabilities). For a pure-state density matrix $\rho_z := |z\rangle\langle z|/\|z\|^2$, $\mathcal{P}(\rho_z) = \sum_s p_s |s\rangle\langle s| = \mathrm{diag}(p)$ with $p_s = |z_s|^2/\|z\|^2$.

**Theorem 4.5 (Pure-state density-matrix decoherence).** Define the **pure-state apex generator** on $z \in \Omega_N \cap S^{2N-1}$ by

$$\mathcal{M}^{\mathrm{ps}}_{\rho_z}(\beta, c) := \mathcal{M}_z(\beta, c)$$

(the lift of the complex apex through the canonical map $z \mapsto \rho_z$ on the amplitude sphere, well-defined up to global phase by Lemma 3.2(b)). Then for every $z \in \Omega_N$ and every $(\beta, c) \in \mathbb{C} \times \widehat{\mathbb{Z}/N}$,

$$\boxed{\mathcal{M}^{\mathrm{ps}}_{\mathcal{P}(\rho_z)}(\beta, c) = \mathcal{M}_{|z|}(\beta, c) = \mathcal{M}_x(\beta, c)\big|_{x=|z|}.}$$

Equivalently: applying the dephasing channel to a pure state, then evaluating the pure-state apex generator, returns the classical apex of the Born-probability ledger under the square-root section.

**Proof.** Dephasing keeps only the diagonal: $\mathcal{P}(\rho_z) = \mathrm{diag}(p)$ with $p_s = |z_s|^2/\|z\|^2$. A diagonal state $\mathrm{diag}(p)$ is read by the pure-state apex generator through its **canonical real-positive amplitude** $\sqrt{p} = (\sqrt{p_1}, \ldots, \sqrt{p_N}) \in S^{N-1}_+$ — the positive-square-root section of the dephasing channel — so $\mathcal{M}^{\mathrm{ps}}_{\mathrm{diag}(p)}(\beta, c) := \mathcal{M}_{\sqrt{p}}(\beta, c)$. Now $\sqrt{p} = |z|/\|z\|$ lies on the same positive ray as the modulus ledger $|z|$, so by positive-scaling invariance (Lemma 3.2(a)) the two have the same centered log-ledger and hence the same apex generator; and by the real-positive reduction (Proposition 3.5), $\mathcal{M}_{|z|}(\beta, c) = \mathcal{M}_x(\beta, c)$ with $x = |z|$. Combining,

$$\mathcal{M}^{\mathrm{ps}}_{\mathcal{P}(\rho_z)}(\beta, c) = \mathcal{M}_{\sqrt{p}}(\beta, c) = \mathcal{M}_{|z|}(\beta, c) = \mathcal{M}_x(\beta, c)\big|_{x=|z|}. \qquad \square$$

Theorem 4.5 is the rank-1 specialization (in the input) of the **mixed-state density-matrix decoherence theorem** scoped below. The pure-state lift $z \mapsto \rho_z$, dephasing $\mathcal{P}$, and positive-square-root section of the

diagonal stratum are precisely the rank-1 inputs to that theorem; Theorem 4.5 therefore closes the abstract claim that *the real-positive sector is the decoherent face of the complex amplitude space* on the pure-state level, with the mixed-state generalization remaining a forward door (§7.2).

**Forward door (mixed-state generalization).** A full **density-matrix apex generator** $\mathcal{M}_\rho(\beta, c)$ generalizing $\mathcal{M}_z$ to mixed states would naturally be defined via the centered modular Hamiltonian

$$L_\rho := \log\rho - \tfrac{1}{N}\mathrm{tr}(\log\rho)\, I_N, \qquad \mathcal{M}_\rho(\beta, c) := \tfrac{1}{N}\mathrm{tr}(e^{\beta L_\rho} C_c^*), \qquad C_c := \sum_s \chi_c(s)|s\rangle\langle s|,$$

with the matrix-logarithm requiring $\rho$ to be full-rank (an obstruction at the pure-state boundary, requiring regularization $\rho \to \rho_\epsilon := (1-\epsilon)\rho + \epsilon I_N/N$ and a limit $\epsilon \to 0$). For diagonal $\rho = \mathrm{diag}(p)$, this reduces (after the half-factor reparameterization $\beta_\rho = \beta/2$ reflecting the amplitude-vs-probability convention noted in the Notation dictionary) to ל0's $\mathcal{M}_x(\beta_\rho, c)$ with $x = \sqrt{p}$. The full mixed-state **density-matrix decoherence theorem** would state: $\mathcal{M}_{\mathcal{P}(\rho)}(\beta, c)$ equals the modulus-restricted classical apex applied to the Born-probability ledger. For *non-diagonal* $\rho$, however, $\log\rho$ and the character operators $C_c$ do not commute, so the $\beta$-Taylor coefficients of $\mathcal{M}_\rho$ are non-commutative moments requiring a choice of operator ordering — and that choice is exactly a choice of quantum monotone (Markov-invariant) metric. Unlike the classical case, Chentsov uniqueness fails and an entire family appears (Morozova–Chentsov 1990; Petz 1996); the dually-flat / convex-potential structure of ל0 §2.4 survives quantization for a *unique* member — the **Kubo–Mori** metric (Grasselli–Streater 2001), not Bures. We develop this fork in §7.2.

Forward direction (planned): **The Monotone-Metric Apex — the Petz Fork and the Kubo–Mori Selection.**

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## 5. The Kummer-cover theorem at integer $\beta$

This section makes the cover-structure preview of Proposition 3.8 into a literal Galois statement at integer $\beta = m \geq 1$. The closure diagonal $\mathcal{M}_z(m, c)$ at integer $m$ is shown to be a section of a line bundle on the projective space $\mathbb{CP}^{N-1}$, associated to a specific character of the **Kummer cover deck group** $\mu_N$; we identify this line bundle explicitly as a power of the tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^{N-1}$. On the real-positive sector the line bundle trivializes (recovering ל0 Proposition 3.7); on the complex sector the deck group acts genuinely. This is the literal complex-side enrichment of ל0 §3.4.

### 5.1 The Kummer cover $\widetilde{\mathbb{CP}^{N-1}} \to \mathbb{CP}^{N-1}$

Recall from ל0 Theorem 3.5 that the closure diagonal at integer $m \geq 1$ takes the form

$$\mathcal{M}_z(m, c) = G(z)^{-m}\widehat{z^m}_c, \qquad G(z) := \Big(\prod_{j=1}^N z_j\Big)^{1/N},$$

where $\widehat{z^m}_c = \frac{1}{N}\sum_s z_s^m \overline{\chi_c(s)}$ is the $c$-th Fourier coefficient of the componentwise $m$-th power, and $G(z)$ is the **complex geometric mean**, defined as an $N$-th root of the product $\prod_j z_j$. The product $\prod_j z_j$ is a single-valued holomorphic function on $\mathbb{C}^N$; the $N$-th root requires a branch choice and is multi-valued by $\mu_N$.

**Definition 5.1 (Kummer cover).** The **Kummer cover** of $\Omega_N \cap \mathbb{CP}^{N-1}$ at level $N$ is the locus

$$\widetilde{\Omega}_N := \{(z, \omega) \in \Omega_N \times \mathbb{C}^\times : \omega^N = \prod_{j=1}^N z_j\} \Big/ \mathbb{C}^\times,$$

where $\mathbb{C}^\times$ acts diagonally $(z,\omega)\mapsto(\lambda z,\lambda\omega)$. The projection $(z,\omega)\mapsto[z]$ realizes $\widetilde{\Omega}_N\to\Omega_N\cap\mathbb{CP}^{N-1}$ as an unramified Galois cover of degree $N$ with deck group

$$\mathrm{Gal}(\widetilde{\Omega}_N\to\Omega_N\cap\mathbb{CP}^{N-1})=\mu_N,$$

acting by $(z,\omega)\mapsto(z,\zeta\omega)$ for $\zeta\in\mu_N$.

On $\widetilde{\Omega}_N$, the geometric mean $G(z)=\omega$ is a **single-valued holomorphic function**; in particular $G(z)^{-m}=\omega^{-m}$ is single-valued on the cover.

**Remark 5.2 (Cover degree refinement).** For the closure diagonal $\mathcal{M}_z(m,c)$ specifically, only the $m$-th power $G(z)^{-m}=\omega^{-m}$ is needed. This is single-valued on the smaller cover $\widetilde{\Omega}_N^{(m)}\to\Omega_N\cap\mathbb{CP}^{N-1}$ of degree $N/\gcd(N,m)$, with deck group $\mu_{N/\gcd(N,m)}$ — the quotient of $\mu_N$ by the kernel $\mu_{\gcd(N,m)}$ of $\zeta\mapsto\zeta^m$. We work with the universal cover $\widetilde{\Omega}_N$ at degree $N$ throughout to keep the statements uniform across $m$.

### 5.2 The Kummer-cover theorem

**Theorem 5.3 (Kummer cover at integer $\beta$).** Fix an integer $m\geq 1$ and a character index $c\in\widehat{\mathbb{Z}/N}$. The complex apex generator $\mathcal{M}_z(m,c)=\omega^{-m}\widehat{z^m}_c$ lifts to a holomorphic function on the Kummer cover $\widetilde{\Omega}_N$, and under the deck-group action $(z,\omega)\mapsto(z,\zeta\omega)$ for $\zeta\in\mu_N$, it transforms by the character

$$\boxed{\mathcal{M}_z(m,c)\longmapsto\rho_m(\zeta)\cdot\mathcal{M}_z(m,c),\qquad\rho_m(\zeta):=\zeta^{-m}.}$$

In particular:

(a) *(Single-valuedness criterion.)* $\mathcal{M}_z(m,c)$ descends to a single-valued function on $\mathbb{CP}^{N-1}$ if and only if $\rho_m$ is the trivial character of $\mu_N$, i.e., $N\mid m$.

(b) *(Line-bundle identification.)* For general integer $m$, $\mathcal{M}_z(m,c)$ is a holomorphic section of the line bundle $\mathcal{L}_m$ on $\Omega_N\cap\mathbb{CP}^{N-1}$ whose total space is

$$\mathcal{L}_m:=\widetilde{\Omega}_N\times_{\mu_N}\mathbb{C}_{\rho_m},$$

where $\mu_N$ acts on the product by $\zeta\cdot(p,v)=(\zeta\cdot p,\rho_m(\zeta)^{-1}v)=(\zeta\cdot p,\zeta^m v)$. Concretely, since $\prod_j z_j$ is a section of $\mathcal{O}(N)$ and its $N$-th root $\omega$ behaves as a section of $\mathcal{O}(1)$, the factor $\omega^{-m}$ is a section of $\mathcal{O}(1)^{\otimes(-m)}=\mathcal{O}(-m)$; equivalently

$$\mathcal{L}_m\;\cong\;\mathcal{O}(N)^{\otimes(-m/N)}\quad\text{as a }\mathbb{Q}\text{-(root-stack) line bundle},\qquad c_1(\mathcal{L}_m)=-m\,H\;\;\text{(integral)},$$

where $H$ is the hyperplane class. The $-m/N$ is the exponent on $\mathcal{O}(N)$, *not* a fractional multiple of $H$: the Chern class is the integer $-mH$ for every $m$. The obstruction to $\mathcal{L}_m$ descending to an honest line bundle on $\mathbb{CP}^{N-1}$ when $N\nmid m$ is the **flat monodromy character** $\rho_m$ of the cyclic cover — a torsion/flat datum invisible to $c_1$ — and descent succeeds iff $\rho_m$ is trivial iff $N\mid m$, in which case $\mathcal{L}_m\cong\mathcal{O}(-m)$.

(c) *(Character independence in c.)* The deck-group action multiplies $\mathcal{M}_z(m,c)$ by the same scalar $\rho_m(\zeta)$ for every character index $c$; the family $\{\mathcal{M}_z(m,c):c\in\widehat{\mathbb{Z}/N}\}$ transforms uniformly under the deck group, hence lies in a single $\mathcal{L}_m$-isotypic component of holomorphic functions on $\widetilde{\Omega}_N$.

**Proof.** On $\widetilde{\Omega}_N$, $G(z)=\omega$ is single-valued. Hence $\mathcal{M}_z(m,c)=\omega^{-m}\widehat{z^m}_c$ is single-valued on $\widetilde{\Omega}_N$ as a product of two single-valued factors ($\omega^{-m}$ on the cover, $\widehat{z^m}_c$ pulled back from $\mathbb{CP}^{N-1}$).

Under deck action $\zeta\in\mu_N$: $\omega\mapsto\zeta\omega$, so $\omega^{-m}\mapsto\zeta^{-m}\omega^{-m}=\rho_m(\zeta)\omega^{-m}$. The Fourier coefficient $\widehat{z^m}_c$ depends only on $z$ (not on $\omega$), so is invariant. Combining,

$$\mathcal{M}_z(m,c)=\omega^{-m}\widehat{z^m}_c\longmapsto\rho_m(\zeta)\,\omega^{-m}\widehat{z^m}_c=\rho_m(\zeta)\mathcal{M}_z(m,c),$$

proving the boxed transformation law.

(a) $\rho_m$ trivial iff $\zeta^{-m} = 1$ for every $\zeta \in \mu_N$ iff $N \mid m$. (b) The associated-bundle construction $\mathcal{L}_m = \widetilde{\Omega}_N \times_{\mu_N} \mathbb{C}_{\rho_m}$ is the standard descent of a $\mu_N$-equivariant trivial bundle on the cover to a line bundle on the base; a function $f$ on $\widetilde{\Omega}_N$ transforming under deck action by $f \mapsto \rho_m(\zeta)f$ is by definition a section of $\mathcal{L}_m$. For the Chern class: $\prod_j z_j$ is a homogeneous form of degree $N$, hence a section of $\mathcal{O}(N)$ with divisor $\sum_j H_j$ of class $NH$. Its $N$-th root $\omega$ is therefore a section of $\mathcal{O}(N)^{1/N} = \mathcal{O}(1)$ (single-valued on the cover), and $\omega^{-m}$ is a section of $\mathcal{O}(-m)$ with $c_1 = -mH$. This integer is the Chern class of $\mathcal{L}_m$ for every $m$; the $-m/N$ that appears is the exponent realizing $\mathcal{L}_m \cong \mathcal{O}(N)^{\otimes(-m/N)}$ as a $\mathbb{Q}$-line bundle, *not* a fractional Chern class. When $N \nmid m$, $\mathcal{L}_m$ fails to descend to $\mathbb{CP}^{N-1}$ because the flat monodromy character $\rho_m$ is non-trivial — a flat obstruction invisible to $c_1$ — and it becomes the honest line bundle $\mathcal{O}(-m)$ exactly when $N \mid m$. (c) is immediate from the proof since $\rho_m(\zeta)$ is independent of $c$. □

**Corollary 5.4 (Eigenspace decomposition).** The space of holomorphic functions on $\widetilde{\Omega}_N$ decomposes by deck-group character as a direct sum of $\rho$-eigenspaces over $\rho \in \widehat{\mu_N} = \widehat{\mathbb{Z}/N}$. The integer-$\beta$ apex generators $\mathcal{M}_z(m, c)$ for $m \in \mathbb{Z}_{\geq 0}$ and $c \in \widehat{\mathbb{Z}/N}$ populate this decomposition by the rule $m \mapsto \rho_m = \chi_{-m \bmod N}$, i.e., the **integer exponent** $m$ **modulo** $N$ is the deck-group character index of the closure-diagonal apex.

### 5.3 The deck-group character vs. the label-character index $c$

A subtlety worth emphasizing: §7's analysis identifies *two* distinct characters of $\mathbb{Z}/N$, both formally elements of $\widehat{\mathbb{Z}/N}$, playing different structural roles in the closure diagonal $\mathcal{M}_z(m, c)$:

1. **Deck-group character** $\rho_m = \chi_{-m \bmod N} \in \widehat{\mu_N}$. Indexes the eigenspace of the Kummer-cover deck-group action; depends on the integer exponent $m$ but is *uniform* across all $c$. This is the "global twist" of the apex generator under the cover.

2. **Label-character index** $c \in \widehat{\mathbb{Z}/N}$. Indexes the eigenspace of the *label* group $\mathbb{Z}/N$ action on the simplex coordinates (the categorical structure); via §0 Theorem 3.5, $\mathcal{M}_z(m, c)$ records the multi-index convolutional sum over $\{(a_1, \ldots, a_m) : a_1 + \cdots + a_m \equiv c \pmod N\}$. This is the "selection rule" of the apex generator.

The two characters live in formally-identical groups $\widehat{\mathbb{Z}/N} \cong \widehat{\mu_N} \cong \mathbb{Z}/N$ but act on different objects:

- $\rho_m$ acts on the $\omega$-coordinate of the cover (the geometric-mean branch).
- $\chi_c$ acts on the $s$-coordinate of the labels (the categorical structure of $\mathbb{Z}/N$).

The cover-character $\rho_m$ is determined by the *exponent* $m$; the label-character $\chi_c$ is determined by the *conductor* $c$ in the moment/closure tower. The fact that both are characters of the same abstract group is a structural coincidence of the categorical-$\mathbb{Z}/N$ setting; in more general label groups, the two characters would live in different dual groups.

**Theorem 5.5 (§0 Theorem 3.5 unaffected by Kummer cover).** The selection rule $a_1 + \cdots + a_m \equiv c \pmod N$ for the convolutional Fourier expansion of $\widehat{z^m}_c$ is *invariant* under the Kummer-cover deck-group action of $\mu_N$. In particular, the deck-group action does not mix character-index $c$ components of $\mathcal{M}_z(m, c)$: it acts by scalar multiplication within each $c$-component independently.

**Proof.** $\widehat{z^m}_c$ is a polynomial in $z = (z_1, \ldots, z_N)$ (the $m$-th power of $z_s$, summed against $\overline{\chi_c(s)}$); the deck-group action $\omega \mapsto \zeta\omega$ does not change $z$, so $\widehat{z^m}_c$ is fixed termwise under the action. The selection-rule expansion (§0 Theorem 3.5 proof) is therefore unaffected. □

### 5.4 Real-positive trivialization and reconciliation with §0 Proposition 3.7

§0 Proposition 3.7 established that on the real-positive sector $\mathbb{R}^N_{>0}$ the $n$-th-power closure $\pi_n$ is a real-analytic homeomorphism, with the Kummer-cover deck group trivializing because $\mu_n \cap \mathbb{R}_{>0} = \{1\}$.

The §7 framework recovers this trivialization as a special case of Theorem 5.3:

**Proposition 5.6 (Real-positive trivialization).** The Kummer cover $\widetilde{\Omega}_N \to \Omega_N \cap \mathbb{CP}^{N-1}$ admits a canonical section over the real-positive sector $\mathbb{R}^N_{>0} \cap \mathbb{CP}^{N-1}$, given by $z \mapsto (z, G_{\mathbb{R}_{>0}}(z))$ where $G_{\mathbb{R}_{>0}}(z) = \left(\prod_j z_j\right)^{1/N} > 0$ is the unique positive real $N$-th root. Restricted to this section, the Kummer-cover line bundle $\mathcal{L}_m$ of Theorem 5.3(b) is canonically trivial, and the deck-group action on $\mathcal{M}_z(m,c)\big|_{\mathbb{R}^N_{>0}}$ reduces to the identity.

**Proof.** On $\mathbb{R}^N_{>0}$, $\prod_j z_j > 0$ and the function $w \mapsto w^{1/N}$ is the unique positive real $N$-th root: a single-valued, real-analytic, positive-real-valued branch. This selects a section of $\widetilde{\Omega}_N \to \Omega_N \cap \mathbb{CP}^{N-1}$ over $\mathbb{R}^N_{>0}$. The line bundle of Theorem 5.3(b) pulled back along this section is trivial because the deck-group orbit through any point in the section consists of $N$ distinct points $(z, \zeta\omega)$ for $\zeta \in \mu_N$, of which only $\zeta = 1$ stays in the real-positive section; hence the section selects one branch of the cover canonically, with trivial deck-group action on the apex. □

Proposition 5.6 makes precise the slogan of ל0 Proposition 3.7: **on the real-positive sector the Kummer cover is a trivial bundle (canonically split by the positive real $N$-th-root section), while on the complex sector $\Omega_N \setminus (U(1) \cdot \mathbb{R}^N_{>0})$ the cover acquires genuine $\mu_N$-deck-group action via Theorem 5.3.**

### 5.5 The complex enrichment: ל0 §3.4 made literal

ל0 §3.4 phrased the Kummer-cover content of the apex generator as an **organizing structure**, with the technical statement (Proposition 3.7) proving only the real-sector trivialization. The non-trivial Galois-action statement was scoped as a forward door to ל7. The forward door is now closed:

> **Theorem 5.3 elevates ל0 §3.4's "Kummer-cover organizing principle" from a real-sector trivialization plus complex-sector conjecture to a complete description: the closure diagonal $\mathcal{M}_z(m,c)$ is a section of the line bundle $\mathcal{L}_m \cong \mathcal{O}(N)^{\otimes(-m/N)}$ (integral Chern class $c_1 = -mH$) on $\mathbb{CP}^{N-1}$, pulled back from the universal degree-$N$ Kummer cover $\widetilde{\mathbb{CP}^{N-1}} \to \mathbb{CP}^{N-1}$ on which the geometric mean $G(z)$ is single-valued. The deck group $\mu_N$ acts by the character $\rho_m(\zeta) = \zeta^{-m}$; the obstruction to descent is the flat character $\rho_m$ (invisible to $c_1$), trivial iff $N \mid m$ (in which case $\mathcal{L}_m \cong \mathcal{O}(-m)$ and $\mathcal{M}_z(m,c)$ descends to $\mathbb{CP}^{N-1}$ unambiguously).**

The "Galois-dual operations" forward door of ל0 §3.4 — pairing the moment diagonal $\partial^n_\beta \mathcal{M}(0,c)$ with the closure diagonal $\mathcal{M}(m,c)$ via the Kummer cover — now has its closure-side fully described. The moment-diagonal side is more delicate (the Taylor expansion at $\beta = 0$ involves the complex log $\log z_s$ at all orders, not just integer powers) and remains a forward door for non-integer $\beta$; we record this in §7.

**Worked example: $N = 2$ (qubit, preview of §6).** For $N = 2$, the Kummer cover $\widetilde{\Omega}_2 \to \Omega_2 \cap \mathbb{CP}^1$ has degree 2, deck group $\mu_2 = \{\pm 1\}$. The closure diagonal at integer $m$: - $m$ even: $\rho_m = \chi_{-m \bmod 2} = \chi_0$ (trivial), so $\mathcal{M}_z(m,c)$ is single-valued on $\mathbb{CP}^1$. - $m$ odd: $\rho_m = \chi_{-1 \bmod 2} = \chi_1$ (the sign character), so $\mathcal{M}_z(m,c)$ is double-valued on $\mathbb{CP}^1$, single-valued on the double cover.

The qubit case at $m = 2$ (Born / RMS rung): single-valued on $\mathbb{CP}^1$ (the Bloch sphere), consistent with the central role of Born probabilities at this rung in ל1 and ל6. The qubit case at $m = 1$ (linear): double-valued on the Bloch sphere — the standard observation that quantum amplitudes carry a $\pm$ sign ambiguity beyond probabilities. *We note that this double-valuedness is the Kummer-cover obstruction in $\sqrt{z_0 z_1}$, whose numerical doubling coincides with — but is logically independent of — the $SU(2) \to SO(3)$ double cover discussed in §2.2: at $N = 2$ both produce a $\mathbb{Z}/2$ obstruction in the amplitude → probability descent, but the Kummer cover persists for all $N$ while the $SU(2) \to SO(3)$ identification does not.* The full Bloch-sphere computation is the subject of §6.

### 5.6 Carlson rigidity: the closure diagonal determines the moment diagonal

The two sampling diagonals of ℵ0 — the moment diagonal (Taylor jets at $\beta = 0$) and the closure diagonal (integer samples $\beta = m$) — were presented there as paired-but-distinct regimes that *happen* to share the modular selection rule $a_1 + \cdots + a_n \equiv c \pmod N$. On the decoherent face the sharing is not a coincidence: it is forced by a classical uniqueness theorem for entire functions of exponential type, and its breakdown is exactly the Kummer regime of this section.

**Proposition 5.7 (Carlson rigidity and its phase threshold).** Fix $c \in \widehat{\mathbb{Z}/N}$.

(a) *(Decoherent face.)* For $x \in \mathbb{R}^N_{>0}$, the entire function $\beta \mapsto \mathcal{M}_x(\beta, c)$ satisfies $|\mathcal{M}_x(it, c)| \le 1$ for every real $t$, hence has exponential type 0 on the imaginary axis. By **Carlson's theorem** (Carlson 1914; Boas 1954, Thm 9.2.1), it is the *unique* entire function of imaginary-axis type $< \pi$ taking the prescribed values $\{\mathcal{M}_x(m, c)\}_{m\ge 0}$ on the non-negative integers. Hence the **closure diagonal** $\{\mathcal{M}_x(m, c)\}_{m\ge 0}$ **determines** $\mathcal{M}_x(\cdot, c)$ **entirely**, and therefore — via the Taylor expansion at $\beta = 0$ — the full moment diagonal $\{\partial^n_\beta \mathcal{M}_x(0, c)\}_{n\ge 0}$.

(b) *(General branch.)* For $z \in \Omega_N$ on a fixed branch, write the centered-argument indicators $\tau_+ := \max_s(-y_s^{(\arg z)})$ and $\tau_- := \max_s(y_s^{(\arg z)})$. Then $\beta \mapsto \mathcal{M}_z(\beta, c)$ has imaginary-axis exponential indicators $(\tau_+, \tau_-)$, and Carlson uniqueness holds iff

$$\tau_+ + \tau_- = \max_s y_s^{(\arg z)} - \min_s y_s^{(\arg z)} < 2\pi,$$

i.e., iff the **centered-argument range is sub-critical**. On the principal branch ($\arg z_s \in (-\pi, \pi]$) this range is always $< 2\pi$, so closure-diagonal rigidity holds *sheet by sheet*.

**Proof.** (a) For real $x$, every $y_s \in \mathbb{R}$, so $|\mathcal{M}_x(it, c)| = \left|\frac{1}{N}\sum_s e^{ity_s}\overline{\chi_c(s)}\right| \le \frac{1}{N}\sum_s 1 = 1$; the imaginary-axis indicator is $0 < \pi$. Carlson's theorem then gives uniqueness from the non-negative-integer samples; the Taylor coefficients at $\beta = 0$ are recovered from the entire function, yielding the moment diagonal. (b) For $z$ on a fixed branch, $y_s = y_s^{(|z|)} + iy_s^{(\arg z)}$, so on $\beta = it$, $|e^{ity_s}| = e^{-ty_s^{(\arg z)}}$; the growth rate as $t \to +\infty$ is $\max_s(-y_s^{(\arg z)}) = \tau_+$ and as $t \to -\infty$ is $\max_s(y_s^{(\arg z)}) = \tau_-$. Carlson's condition $\tau_+ + \tau_- < 2\pi$ is the stated range bound, and $\arg z_s \in (-\pi, \pi]$ forces range $< 2\pi$ on the principal branch. □

**Remark 5.8 (Carlson rigidity and the Kummer cover are the same threshold).** Proposition 5.7 explains, on the decoherent face, *why* the two diagonals share the selection rule of ℵ0 Theorems 3.3 and 3.5: they are not independent samples but two reads of a single function that its integer samples already over-determine. The rigidity is **per sheet**. Branch shifts $\log z_s \mapsto \log z_s + 2\pi i k_s$ (Proposition 3.8) translate $y_s^{(\arg z)}$ by $2\pi(k_s - \bar{k})$, pushing the centered-argument range across the $2\pi$ threshold and breaking *cross-sheet* determination; the cross-sheet ambiguity is exactly the $\mu_N$ deck-group action of Theorem 5.3. Thus **Carlson rigidity (within a sheet) and the Kummer cover (between sheets) are two faces of the same $2\pi$-threshold**: the closure diagonal determines the generator on each Kummer sheet, and the deck group $\mu_N$ is precisely the obstruction to gluing those determinations into a single-valued function on $\mathbb{CP}^{N-1}$ when $N \nmid m$. The Wick-rotation reading — replicator gradient flow on real $\beta$ (ℵ0 §2.4) versus the Lee–Yang lattice on imaginary $\beta$ (Proposition 6.2) — places both diagonals on one holomorphic plane (§7.4); Carlson rigidity is the statement that the real-integer samples already fix the whole plane, branch by branch.

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## 6. The qubit anchor: $N = 2$ in full

We compute the complex apex generator $\mathcal{M}_z(\beta, c)$ in full at $N = 2$ — the smallest non-trivial dimension, the standard qubit of quantum information. The Bloch sphere $\mathbb{CP}^1 \cong S^2$ provides explicit coordinates; the decoherent equator (real-positive sector) recovers ℵ0's qubit apex; the off-equatorial complex sector exhibits the phase deformation of Proposition 4.3 in closed form; the Kummer cover of §5 acts as the $\mu_2 = \{\pm 1\}$

deck group, splitting the integer-$\beta$ rungs into single-valued ($m$ even) and double-valued ($m$ odd) families; and ל0 Proposition 2.5's Lee–Yang lattice lifts to a **phase-twisted lattice** that rotates and scales as one moves off the decoherent equator.

### 6.1 The Bloch sphere parameterization

A normalized qubit amplitude $z \in S^3 \subset \mathbb{C}^2$, modulo global phase, has the standard Bloch parameterization

$$z = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi}\sin(\theta/2) \end{pmatrix}, \qquad \theta \in [0,\pi],\ \phi \in [0,2\pi),$$

covering $\mathbb{CP}^1 \cong S^2$ as $(\theta,\phi)$ range over standard spherical coordinates. The corresponding **Bloch vector** in $\mathbb{R}^3$ is

$$\vec{r}(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta),$$

with the north pole $\theta = 0$ corresponding to $|0\rangle$, the south pole $\theta = \pi$ to $|1\rangle$, and the equator $\theta = \pi/2$ to equal-modulus superpositions of $|0\rangle$ and $|1\rangle$ with phase $\phi$. The Born probabilities are

$$p_0 = \cos^2(\theta/2) = \tfrac{1+\cos\theta}{2}, \qquad p_1 = \sin^2(\theta/2) = \tfrac{1-\cos\theta}{2}.$$

The **decoherent equator** is the meridian $\phi = 0$, the real-positive sector $\mathbb{R}^2_{>0} \cap S^3$ of ל1–ל6. Points off this meridian carry non-trivial phase and exhibit the complex apex structure of ל7.

### 6.2 The complex apex on the qubit in closed form

For $N = 2$, the centered log-ledger is one-dimensional: $y_0 = -y_1 = \frac{1}{2}\log(z_0/z_1)$. Setting $\eta := y_0$, Definition 3.3 gives

$$\mathcal{M}_z(\beta,0) = \tfrac{1}{2}(e^{\beta\eta} + e^{-\beta\eta}) = \cosh(\beta\eta), \qquad \mathcal{M}_z(\beta,1) = \tfrac{1}{2}(e^{\beta\eta} - e^{-\beta\eta}) = \sinh(\beta\eta),$$

using the two characters $\chi_0 \equiv 1$ and $\chi_1(s) = (-1)^s$ on $\mathbb{Z}/2$.

**Proposition 6.1 (Qubit apex in Bloch coordinates).** In the Bloch parameterization,

$$\eta(\theta,\phi) = \tfrac{1}{2}\log\cot(\theta/2) - \tfrac{i\phi}{2} = y_0^{(|z|)}(\theta) + i y_0^{(\arg z)}(\phi),$$

where $y_0^{(|z|)}(\theta) = \frac{1}{2}\ln\cot(\theta/2)$ is the real modulus log-ledger and $y_0^{(\arg z)}(\phi) = -\phi/2$ is the centered argument. The complex apex on the qubit is therefore

$$\boxed{\mathcal{M}_z(\beta,0) = \cosh\left(\tfrac{\beta}{2}\ln\cot(\theta/2) - \tfrac{i\beta\phi}{2}\right), \qquad \mathcal{M}_z(\beta,1) = \sinh\left(\tfrac{\beta}{2}\ln\cot(\theta/2) - \tfrac{i\beta\phi}{2}\right).}$$

**Proof.** With $z_0 = \cos(\theta/2)$, $z_1 = e^{i\phi}\sin(\theta/2)$,

$$\frac{z_0}{z_1} = \cot(\theta/2)\, e^{-i\phi} \quad \Longrightarrow \quad \log\left(\frac{z_0}{z_1}\right) = \ln\cot(\theta/2) - i\phi,$$

so $\eta = \frac{1}{2}\log(z_0/z_1) = \frac{1}{2}\ln\cot(\theta/2) - \frac{i\phi}{2}$, matching the modulus-phase decomposition of Lemma 3.2(c) at $N = 2$. Substituting into the cosh / sinh forms above yields the boxed expressions. □

### 6.3 The decoherent equator and reduction to ℸ0

**On the equator $\phi = 0$:** $\eta = \frac{1}{2}\ln\cot(\theta/2) \in \mathbb{R}$, and the apex specializes to

$$\mathcal{M}_z(\beta, 0)\big|_{\phi=0} = \cosh(\tfrac{\beta}{2}\ln\cot(\theta/2)) = \tfrac{1}{2}(\cot(\theta/2)^{\beta/2} + \tan(\theta/2)^{\beta/2}),$$

$$\mathcal{M}_z(\beta, 1)\big|_{\phi=0} = \sinh(\tfrac{\beta}{2}\ln\cot(\theta/2)) = \tfrac{1}{2}(\cot(\theta/2)^{\beta/2} - \tan(\theta/2)^{\beta/2}).$$

Comparing with ℸ0 applied to the modulus ledger $|z| = (\cos(\theta/2), \sin(\theta/2))$: the ℸ0 centered log-ledger is $y_0^{(\text{ℸ}0)} = \frac{1}{2}\ln\cot(\theta/2) = \eta\big|_{\phi=0}$, so $\mathcal{M}_x(\beta, c)$ from ℸ0 Definition 3.1 agrees with $\mathcal{M}_z(\beta, c)\big|_{\phi=0}$ — the decoherence theorem (Theorem 4.2) at $N = 2$.

**Born / RMS rung at $\beta = 2$.** Specializing to $\beta = 2$ on the equator,

$$\mathcal{M}_z(2, 0)\big|_{\phi=0} = \tfrac{1}{2}(\cot(\theta/2) + \tan(\theta/2)) = \frac{1}{\sin\theta},$$

$$\mathcal{M}_z(2, 1)\big|_{\phi=0} = \tfrac{1}{2}(\cot(\theta/2) - \tan(\theta/2)) = \frac{\cos\theta}{\sin\theta} = \cot\theta,$$

recovering ℸ6's Born-closure structure at the qubit: in terms of Born probabilities $p_0 = \cos^2(\theta/2)$, $p_1 = \sin^2(\theta/2)$, $\sin\theta = 2\sqrt{p_0 p_1}$ and $\cos\theta = p_0 - p_1$, so $\mathcal{M}_z(2,0)|_{\phi=0} = 1/(2\sqrt{p_0p_1})$ and $\mathcal{M}_z(2,1)|_{\phi=0} = (p_0 - p_1)/(2\sqrt{p_0p_1})$. The non-trivial mode $c = 1$ measures the **probability gap** $p_0 - p_1$ relative to the Born normalization, the qubit signature of conductor-2 structure on $\mathbb{Z}/2$.

### 6.4 The Kummer cover at $N = 2$: $m$-parity selection

For $N = 2$ the Kummer cover (§5) has deck group $\mu_2 = \{\pm 1\}$, and Theorem 5.3 says $\mathcal{M}_z(m, c) \mapsto \rho_m(\zeta)\mathcal{M}_z(m, c)$ with $\rho_m(\zeta) = \zeta^{-m}$.

**At $\zeta = -1$:** $\rho_m(-1) = (-1)^{-m} = (-1)^m$. So the deck-group action multiplies $\mathcal{M}_z(m, c)$ by $(-1)^m$:

- $m$ **even** ($m = 0, 2, 4, \ldots$): $\rho_m$ trivial, $\mathcal{M}_z(m, c)$ single-valued on $\mathbb{CP}^1$.
- $m$ **odd** ($m = 1, 3, 5, \ldots$): $\rho_m$ the sign character, $\mathcal{M}_z(m, c)$ double-valued on $\mathbb{CP}^1$, single-valued on the double cover.

**Worked example: $m = 1$ vs $m = 2$.** Direct from the boxed forms of Proposition 6.1:

| $m$ | $\mathcal{M}_z(m, 0)$ | $\mathcal{M}_z(m, 1)$ | Branch behavior under $\phi \mapsto \phi + 2\pi$ |
|---|---|---|---|
| 1 | $\cosh(\frac{1}{2}\ln\cot(\theta/2) - \frac{i\phi}{2})$ | $\sinh(\frac{1}{2}\ln\cot(\theta/2) - \frac{i\phi}{2})$ | $\to -\mathcal{M}_z(1, c)$ (double-valued) |
| 2 | $\cosh(\ln\cot(\theta/2) - i\phi)$ | $\sinh(\ln\cot(\theta/2) - i\phi)$ | $\to +\mathcal{M}_z(2, c)$ (single-valued) |

The $m = 1$ entries acquire $\cosh(\ldots - i\pi) = -\cosh(\ldots)$ under $\phi \mapsto \phi + 2\pi$ (since $\cosh(z + i\pi) = -\cosh(z)$ and similarly for sinh); the $m = 2$ entries are invariant since $\cosh(\ldots - 2i\pi) = \cosh(\ldots)$. This is the $\pm 1$ **amplitude sign ambiguity** of standard quantum mechanics: at the linear ($m = 1$) rung, the apex is sensitive to the global $\pm$ choice of amplitude — a $\mathbb{Z}/2$ obstruction in the descent from amplitude to probability that coincides numerically with the $SU(2) \to SO(3)$ double cover at $N = 2$ (§2.2) but is structurally the Kummer-cover obstruction of Theorem 5.3 (the Kummer obstruction persists for all $N$; the $SU(2) \to SO(3)$ identification is a low-dimensional coincidence). At the Born ($m = 2$) rung, the apex sees only probability-level data and the obstruction disappears.

**Polynomial-rational closed forms.** Direct from the definition $\mathcal{M}_z(m,c) = G(z)^{-m}\widehat{z^m}_c$:

$$\mathcal{M}_z(1,0) = \frac{z_0+z_1}{2\sqrt{z_0 z_1}}, \qquad \mathcal{M}_z(1,1) = \frac{z_0-z_1}{2\sqrt{z_0 z_1}},$$
$$\mathcal{M}_z(2,0) = \frac{z_0^2+z_1^2}{2z_0 z_1}, \qquad \mathcal{M}_z(2,1) = \frac{z_0^2-z_1^2}{2z_0 z_1}.$$

The $\sqrt{z_0 z_1}$ in the $m=1$ denominator is the source of the $\mu_2$ branch ambiguity; at $m=2$ the square root squares away, and the apex is a **rational function** of $z$, defining a holomorphic function on the open Bloch sphere $\mathbb{CP}^1 \setminus \{|0\rangle, |1\rangle\}$ (the poles at the basis states correspond to one Born probability vanishing, where the apex generator blows up — consistent with §0 Proposition 2.5's Lee–Yang structure at extreme $\theta$).

### 6.5 The phase-twisted Lee–Yang lattice

§0 Proposition 2.5 located the Lee–Yang zeros of $Z_y(\beta) = \cosh(a\beta)$ at $N=2$ on the imaginary $\beta$-lattice $\{i\pi(k+1/2)/a : k \in \mathbb{Z}\}$. §7 lifts this to the full Bloch sphere via the complex parameter $\eta(\theta,\phi) = \frac{1}{2}\ln\cot(\theta/2) - \frac{i\phi}{2}$.

**Proposition 6.2 (Phase-twisted Lee–Yang lattice).** The zero set of $\mathcal{M}_z(\beta,0) = \cosh(\beta\eta)$ in $\beta \in \mathbb{C}$, at the Bloch point $(\theta,\phi)$ with $\eta = \eta(\theta,\phi) \neq 0$, is the lattice

$$\boxed{\{\beta \in \mathbb{C} : \beta\eta = i\pi(k+\tfrac{1}{2}),\ k \in \mathbb{Z}\} = \Big\{\beta_k = \frac{i\pi(k+\frac{1}{2})}{\eta(\theta,\phi)} : k \in \mathbb{Z}\Big\}.}$$

Writing $\eta = ae^{-i\psi}$ in polar form (so $a = |\eta| > 0$ and $\psi = -\arg\eta \in (-\pi,\pi]$), the lattice becomes

$$\beta_k = \frac{\pi(k+\frac{1}{2})}{a} \cdot e^{i(\pi/2+\psi)},$$

a uniformly-spaced lattice along the ray of angle $\pi/2+\psi$ in $\mathbb{C}$ with spacing $\pi/a$. On the decoherent equator $\phi = 0$: $\psi = 0$, lattice on positive imaginary axis (recovers §0 Proposition 2.5). Off the equator: the lattice **rotates** by $\psi = \arctan(\frac{\phi}{\ln\cot(\theta/2)})$ and **rescales** by $1/a = 1/\sqrt{\frac{1}{4}(\ln\cot(\theta/2))^2 + \frac{1}{4}\phi^2}$.

**Proof.** $\cosh w = 0$ iff $w = i\pi(k+\frac{1}{2})$ for $k \in \mathbb{Z}$ (§0 Proposition 2.5 base case). Setting $w = \beta\eta$ and solving for $\beta$: $\beta = i\pi(k+\frac{1}{2})/\eta$. Polar substitution $\eta = ae^{-i\psi}$ gives $1/\eta = e^{i\psi}/a$, so $\beta_k = i\pi(k+\frac{1}{2})e^{i\psi}/a = \pi(k+\frac{1}{2})e^{i(\pi/2+\psi)}/a$, as stated. □

**Geometric picture.** As the Bloch point moves from the decoherent equator $\phi = 0$ to the "phase-rotated" point $\phi = \pi/2$ (with $\theta$ fixed in $(0,\pi/2)$), the Lee–Yang lattice **continuously rotates** by an angle that grows from 0 at $\phi = 0$ to $\arctan(\pi/(2\ln\cot(\theta/2)))$ at $\phi = \pi/2$. At the **uniform-modulus equator** $\theta = \pi/2$ (where $\ln\cot(\theta/2) = 0$), the rotation angle reaches $\pi/2$ for any $\phi \neq 0$ — the Lee–Yang lattice rotates to the **real $\beta$-axis**, with spacing $2\pi/|\phi|$ (from $a = |\phi|/2$). At the precise point $\theta = \pi/2$, $\phi = 0$ (uniform superposition with zero phase), $\eta = 0$, the lattice degenerates ($Z_y(\beta) \equiv 1$ has no zeros), the Lee–Yang lattice retreats to infinity.

**Lee–Yang condensation interpretation.** The rotation of the Lee–Yang lattice from the imaginary axis (classical / decoherent) toward the real axis (maximally phase-rotated) is the qubit-level analog of **Lee–Yang condensation** in statistical-mechanics terminology: the zeros of the partition function move from the imaginary axis (no genuine singularity in real $\beta$) toward the real positive axis (a genuine vanishing of the apex generator at specific real $\beta$). At the uniform-modulus equator with non-zero $\phi$, the partition function $\cosh(\beta\eta)$ has zeros on the real positive $\beta$-axis precisely at $\beta_k = 2\pi(k+\frac{1}{2})/|\phi|$ — these are not thermodynamic phase transitions in any physical sense, but they are precisely the *quantum-coherence singularities* of the apex generator: real-$\beta$ points at which the modulus-zero phase content produces an exact cancellation between the two basis-state contributions. We flag this as the natural finite-dimensional toy of the Lee–Yang circle theorem (Lee–Yang 1952) for purely-phase ledgers; whether this extends to a general Lee–Yang-style theorem on $\mathcal{M}_z$ for $N \geq 3$ is forward-door content (§7.6).

### 6.6 Qubit summary table

We assemble the qubit content into a single table organized by integer exponent rung $m$ and character index $c \in \{0, 1\}$.

| Rung | $m$ | Apex $\mathcal{M}_z(m, 0)$ | Apex $\mathcal{M}_z(m, 1)$ | Kummer cover | Decoherent ($\phi = 0$) interpretation |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | trivial | normalization |
| Linear | 1 | $(z_0 + z_1)/(2\sqrt{z_0 z_1})$ | $(z_0 - z_1)/(2\sqrt{z_0 z_1})$ | $\mu_2$, double-valued | amplitude sign / $SU(2) \to SO(3)$ |
| Born | 2 | $(z_0^2 + z_1^2)/(2z_0 z_1)$ | $(z_0^2 - z_1^2)/(2z_0 z_1)$ | trivial | $1/\sin\theta$, $\cos\theta/\sin\theta = \cot\theta$ |
| Cubic | 3 | $\cosh(3\eta)$ | $\sinh(3\eta)$ | $\mu_2$, double-valued | ל6 §8 / Waring cubic rung |
| Quartic | 4 | $\cosh(4\eta)$ | $\sinh(4\eta)$ | trivial | $m \equiv 0 \pmod 2$ |
| Generic $\beta$ | — | $\cosh(\beta\eta)$ | $\sinh(\beta\eta)$ | holomorphic on the universal cover | full holomorphic family |

The decoherent equator $\phi = 0$ recovers ל0 (Theorem 4.2); the off-equatorial complex sector exhibits the phase deformation (Proposition 4.3 at $N = 2$) and the phase-twisted Lee–Yang lattice (Proposition 6.2); the Kummer cover (§5) acts uniformly across $c$, splitting the integer rungs into single-valued (even $m$) and double-valued (odd $m$) families.

**$N = 2$ as the structural anchor of ל7.** Just as $N = 6$ is ל0's structural anchor (the smallest cross-prime case), $N = 2$ is ל7's structural anchor: the smallest dimension where the Hopf fibration, Bloch sphere, and $\mu_2$ Kummer cover all appear non-trivially, the qubit of quantum information, the $SU(2) \to SO(3)$ double cover, and the toy Lee–Yang lattice of ל0 Proposition 2.5 all simultaneously. Every claim of ל7's main theorems (the three-layer factorization, Fubini–Study restriction, modulus-phase factorization, decoherence theorem, Kummer-cover theorem) acquires a fully explicit closed form at this anchor.

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## 7. Forward doors from ל7

ל7 closes the complex-amplitude door of ל0 at the pure-state level. Six structural extensions sit naturally beyond it; we catalog them in the corridor format of ל0 §6. The working titles below are deliberately left **unnumbered**: the ל-series slots ל8–ל18 are already occupied by other manuscripts in the programme, and several of these directions overlap existing papers (notably the continuum/Hilbert-space door and the finite-to-Archimedean programme, and the critical-line door and the multiplicative-apex paper ל8). Canonical numbers are to be assigned when each is mapped onto — or split off from — the existing suite.

### 7.1 Multi-particle structure and the entanglement stratum

Tensor products $\mathbb{C}^N \otimes \mathbb{C}^M$ realize multi-particle quantum systems on $NM$ outcomes. The complex apex $\mathcal{M}_z(\beta, c)$ on a product state $z = u \otimes v$ ($u \in \mathbb{C}^N$, $v \in \mathbb{C}^M$) factorizes through the CRT-style product

$$\mathcal{M}_{u\otimes v}(\beta, (c_1, c_2)) = \mathcal{M}_u^{(N)}(\beta, c_1) \cdot \mathcal{M}_v^{(M)}(\beta, c_2) \quad \text{(product-state factorization)}.$$

The **non-factorization** of $\mathcal{M}_z$ on entangled states is therefore a categorical-simplex measurement of entanglement, analogous to ל0 §4's CRT cross-prime interaction stratum but lifted to the genuinely quantum (Hopf-and-toric covered) setting. The Schmidt decomposition $z = \sum_k \sigma_k u_k \otimes v_k$ refines this into a convex-combination expansion of $\mathcal{M}_z$ over the Schmidt rungs, with the Schmidt coefficients $\sigma_k \in \mathbb{R}_{>0}$ playing the role of a probability measure on the rungs.

**Conjecture 7.1 (Entanglement measure from the apex).** There exists an entanglement monotone $E_{\text{apex}}(z)$ on the joint amplitude space $\mathbb{C}^N \otimes \mathbb{C}^M$, defined as the obstruction-norm of $\mathcal{M}_z$ from the product-state factorization above (e.g., the $L^2$ distance over $(\beta, c_1, c_2)$ between $\mathcal{M}_z$ and the closest product), that recovers von-Neumann-entropy-style entanglement on the Schmidt decomposition.

Forward direction (planned forward paper, number to be assigned): **Multi-Particle Tensor Products and the Apex of Entanglement.**

### 7.2 Mixed states, the Petz fork, and the Kubo–Mori selection

Theorem 4.5 settles the pure-state density-matrix decoherence. The mixed-state forward step — the full $\mathcal{M}_\rho(\beta, c)$ via the modular Hamiltonian $L_\rho = \log\rho - \frac{1}{N}\mathrm{tr}(\log\rho)\, I_N$ — exposes a structural fact the pure-state sector hid, and it is the deepest continuation of ל0's founding question.

**The Petz fork.** On the simplex, Fisher–Rao is the *unique* monotone (Markov-invariant) Riemannian metric (Chentsov 1982) — the uniqueness every gauge of ל0 shares at the uniform point (§1.4). On density matrices this uniqueness **fails**: the monotone metrics form an infinite family $g^f$ indexed by operator-monotone $f$ with $f(1) = 1$ and $f(t) = tf(1/t)$, with coherence weight $c_f(x, y) = 1/(y\, f(x/y))$ — the Morozova–Chentsov–Petz classification (Morozova–Chentsov 1990; Petz 1996). Three named rungs, ordered harmonic $\le f \le$ arithmetic (Petz–Sudár 1996):

| metric | $f(t)$ | mean $\mathfrak{m}_f$ | role |
|---|---|---|---|
| Bures / SLD | $(1+t)/2$ | arithmetic | quantum Cramér–Rao (estimation); *minimal* monotone metric |
| Kubo–Mori (BKM) | $(t-1)/\log t$ | logarithmic | Hessian of relative entropy; *dually flat* |
| RLD | $2t/(1+t)$ | harmonic | *maximal* monotone metric |

So a naïve "the quantum Fisher information" is ambiguous; a gauge $f$ must be chosen — the **operator-valued continuation of ל0's scalar power-mean corridor**. The classical question *what is additive?* (a scalar mean, the power $r$) becomes *how are coherences weighted?* (an operator mean, the function $f$). Concretely, for non-diagonal $\rho$ the operators $\log\rho$ and $C_c$ do not commute, so the $\beta$-Taylor coefficients of $\mathcal{M}_\rho$ are non-commutative moments whose operator ordering *is* the choice of $f$.

**Decoherent-face collapse (metric form of the decoherence theorem).** Every monotone metric agrees on the diagonal stratum: for $\rho = \mathrm{diag}(p)$ and a diagonal tangent $V = \mathrm{diag}(v)$ with $\sum_s v_s = 0$,

$$g^f_\rho(V, V) = \sum_s c_f(p_s, p_s)\, v_s^2 = \sum_s \frac{v_s^2}{p_s} = g^{\mathrm{FR}}_p(v, v) \quad \text{for every } f,$$

since $c_f(x, x) = 1/(x\, f(1)) = 1/x$ — the *idempotency of means* $\mathfrak{m}_f(x, x) = x$. This is the metric counterpart of Theorem 4.2: the gauge freedom of the fork vanishes exactly on the real-positive (decoherent) face. Chentsov uniqueness is, in this light, no deep accident — it is the statement that all means agree when their two arguments coincide.

**The apex generator forces Kubo–Mori, not Bures.** Along a single escort line $\rho_\beta = e^{\beta H}/\operatorname{tr} e^{\beta H}$ the generator is gauge-free and reproduces ל0's cumulant tower of $\operatorname{spec}(H)$ (no $f$ appears, since $H$ commutes with $e^{\beta H}$). The fork bites only in the *transverse* second-order structure: for a non-commuting exponential family $\rho_\theta = \exp(\theta^i H_i - \psi(\theta))$, the Hessian $\partial_i \partial_j \psi$ equals the monotone metric $g^f$ **iff** $f$ is Kubo–Mori (Amari–Nagaoka 2000, ch. 7; uniqueness of the dually-flat monotone metric: Grasselli–Streater 2001). Since $\mathcal{M}_\rho$ is an exponential-family object in $L_\rho = \log \rho - \cdots$, it is the convex potential of a dually-flat geometry **only** for Kubo–Mori. Hence the corrected reading: the density-matrix apex is the **Kubo–Mori potential**; **Bures is the estimation member** (quantum Cramér–Rao, Braunstein–Caves 1994) and the pure-state Fubini–Study restriction of §2, agreeing with Kubo–Mori only on the decoherent face. This mirrors the scalar corridor exactly: ל0 already singled out the geometric/log gauge ($r = 0$) as the potential-carrying center; Kubo–Mori — the $\log \rho$ metric — is its operator continuation, while Bures is the analog of the $r = 2$ Born gauge (operationally central, but not potential-carrying).

**A metric for the phase fiber (by-product for ל7).** At $\rho = \operatorname{diag}(p)$ every $g^f$ is diagonal in the $\mathbb{Z}/N$ coherence-order grading $X = \sum_k X^{(k)}$, $X^{(k)} = \sum_{s-s' \equiv k} X_{ss'} |s\rangle\langle s'|$: the order-0 block is the gauge-rigid Fisher–Rao metric of §2, while the order-$k \neq 0$ blocks — exactly the tangent directions of the relative-phase fiber $T^{N-1}$ of §1.2 — carry the *entire* gauge dependence, with block weight $c_f(p_s, p_{s+k})$. The fork thus equips the phase fiber, introduced in §1 only as a symplectic/toric object, with a Riemannian metric for the first time; the coherence order $k$ and the conductor index $c$ are the same $\mathbb{Z}/N$-grading read on operators versus on functions, so the metric is conductor-graded.

**Conjecture 7.2 (Density-matrix decoherence theorem).** Let $\mathcal{P}$ be the computational-basis dephasing channel. Then for every full-rank $\rho$, every $\beta \in \mathbb{C}$, and every $c \in \widehat{\mathbb{Z}/N}$,

$$\mathcal{M}_{\mathcal{P}(\rho)}(\beta, c) = \mathcal{M}_{\sqrt{\operatorname{diag}(\rho)}}(\beta_{\mathrm{ps}} = 2\beta, c),$$

extending Theorem 4.5 from the pure-state stratum to the full density-matrix simplex. (This statement is *metric-independent* — it concerns the generator's value, not its Hessian — so it is unaffected by the Petz fork.)

Forward direction (planned): **The Monotone-Metric Apex — the Petz Fork and the Kubo–Mori Selection** (Bures = SLD/estimation rung; Kubo–Mori = dually-flat/potential rung; the two agree on the decoherent face).

### 7.3 The Kubo–Mori replicator and quantum natural-gradient flow

The replicator $\beta$-flow of ל0 §2.4 is the Shahshahani / Fisher gradient flow on the simplex. Its quantum lift is the natural-gradient flow on the density-matrix manifold along the monotone metric selected in §7.2. Because the free-energy identities $\partial_\beta \Psi = \langle L_\rho \rangle$, $\partial_\beta^2 \Psi = \operatorname{Var}$ of ל0 §2.4 hold only for the dually-flat member, the correct quantum replicator is the **Kubo–Mori** (BKM) flow, not the Bures one. The complex apex $\mathcal{M}_z(\beta, c)$ remains the **free energy** along this flow on each character mode; on a single escort line it is gauge-free (§7.2).

**Conjecture 7.3 (Kubo–Mori replicator).** The escort family $\rho_\beta := e^{\beta L_\rho}/\operatorname{tr}(e^{\beta L_\rho})$ on density matrices is the Kubo–Mori natural-gradient flow of the expectation functional $\rho \mapsto \operatorname{tr}(\rho L_\rho)$, and $\partial_\beta \mathcal{M}_\rho(\beta, c)$ is the BKM (Bogoliubov–Kubo–Mori) covariance of $L_\rho$ against the character operators $C_c$ — the quantum analog of the replicator/character-convolution identity of ל0 §2.4. The Bures natural-gradient flow is the distinct *estimation-side* companion, coinciding with the Kubo–Mori flow only on the decoherent face.

Forward direction (planned forward paper, number to be assigned): **The Kubo–Mori Replicator and Quantum Natural-Gradient Flow** (with the Bures flow as the estimation-side companion).

### 7.4 Holomorphic flow and the zero structure of $\mathcal{M}_z$

Proposition 3.4 establishes $\mathcal{M}_z(\beta, c)$ as entire in $\beta$ on each branch. The qubit case (Proposition 6.2) gives the complete zero set in closed form: a phase-twisted lattice that rotates from imaginary to real as the Bloch point moves off the decoherent equator. For $N \geq 3$, the zero set is not pinned to any single curve by elementary arithmetic; its global structure is the natural finite-categorical analog of the zeros of an $L$-function.

Two structural facts already organize this holomorphic plane. First, **Wick rotation**: the replicator gradient flow of ל0 §2.4 lives on the *real* $\beta$-axis (dissipative ascent of mean fitness), while the Lee–Yang zeros sit on the *imaginary* axis (Proposition 6.2); $\beta \mapsto i\beta$ rotates the dissipative replicator into oscillatory unitary evolution and marks the quantum resonances at the Lee–Yang lattice. Second, **Carlson rigidity** (Proposition 5.7) is already a theorem, not a door: on the decoherent face the real-integer closure samples over-determine the entire generator, so the two diagonals are rigidly linked, and the rigidity degrades sheet-to-sheet exactly through the $\mu_N$ Kummer cover (Remark 5.8). The forward content of this door is therefore the *complex-stratum* zero geometry for $N \geq 3$, where phase winding past the Carlson threshold and the Kummer monodromy jointly shape the zero variety.

**Conjecture 7.4 (Lee–Yang on the complex apex).** For ledgers $z$ in suitable arithmetic strata of $\mathbb{C}^N$ (e.g., when $|z_s|$ encodes prime structure on $\mathbb{Z}/N$), the zeros of $\mathcal{M}_z(\beta, c)$ in $\beta$ lie on a structured subvariety of $\mathbb{C}$ that mixes imaginary-axis content (the decoherent / Lee–Yang content of ל0) with off-axis phase-twisted content (the Bloch-rotated content of Proposition 6.2).

Forward direction (planned forward paper, number to be assigned): **The Lee–Yang Zero Variety of the Complex Apex.**

### 7.5 Infinite-dimensional limit ($N \to \infty$, Hilbert-space apex)

The complex amplitude space $\mathbb{C}^N$ specializes finite-dimensional quantum mechanics. The continuum / Hilbert-space limit replaces $\mathbb{Z}/N$ by a continuous label group (the real line, the circle, a Lie group), $\mathbb{C}^N$ by a separable Hilbert space, and the moment map $\pi_{\text{Born}}$ by the Born measure of a quantum state. The complex apex $\mathcal{M}_z(\beta, c)$ should generalize to an integral kernel $\mathcal{M}_\psi(\beta, \xi)$ depending on a Mellin parameter $\beta \in \mathbb{C}$ and a Pontryagin dual $\xi \in \widehat{G}$.

**Conjecture 7.5 (Hilbert-space apex).** The complex apex of a Hilbert-space state $|\psi\rangle$ on $G$ is the Mellin transform of the centered log-amplitude under the Pontryagin dual character $\chi_\xi$, holomorphic in $\beta$ on appropriate functional-analytic domains, with the decoherence theorem extending to the modulus-restriction $\psi \mapsto |\psi|$ in the position representation. The finite-categorical ל7 is the $G = \mathbb{Z}/N$ specialization.

Forward direction (planned forward paper, number to be assigned; cf. the existing finite-to-Archimedean completion programme): **Hilbert-Space Apex and Continuum Quantum Sectors.**

### 7.6 RH-shaped questions at the complex apex

ל0 §6.6 / §6.7 asks whether the zeros of the apex generator line up on a critical line for arithmetically structured ledgers. The complex extension provides richer phase content: the cross-prime interaction stratum on $\mathbb{Z}/N$ for $N$ with multiple prime factors acquires the Hopf-and-toric phase structure, and the additive RH-shaped question (do zeros of $\mathcal{M}_z$ for primorial ledgers concentrate on a critical line in $\beta \in \mathbb{C}$?) becomes phase-sensitive in a precise way.

**Aspiration vs. theorem.** Posed over the *additive* characters $\chi_c(s) = e^{2\pi i c s/N}$ of $\mathbb{Z}/N$, this is an aspiration, not a concrete target, and we record it as such. The additive apex $\mathcal{M}_z(\beta, c)$ is a finite exponential sum in $\beta$; its zero set is a phase-twisted lattice (Proposition 6.2) carrying no arithmetic normalization that would *force* the zeros onto a line. There is no Frobenius, no cohomological weight, and no $\sqrt{q}$ scale in the additive $\mathbb{Z}/N$ setting: $\mathbb{Z}/N$ is the residue ring of the number field $\mathbb{Q}$, and the matching critical-line statement there

(RH for $\zeta$, GRH for Dirichlet $L$) is itself the open conjecture. Conjecture 7.6 is therefore a heuristic; the *concrete provable* RH target is relocated to the multiplicative / function-field apex of ל8 described below, where the analogous statement is a theorem of Weil.

**Conjecture 7.6 (Complex critical line — additive aspiration).** For primorial $N = P_k = \prod_{p \le p_k} p$ and ledger $x$ encoding prime structure on $\mathbb{Z}/N$ (e.g., the indicator of units in $(\mathbb{Z}/N)^\times$), the zeros of $\mathcal{M}_z(\beta, c)$ in $\beta \in \mathbb{C}$ for $z$ on the *decoherent equator* of the complex extension concentrate on a critical line $\mathrm{Re}\,\beta = \beta_*$ in a finite-categorical analog of RH; the off-equator phase-twisted zeros rotate the critical line by a phase angle controlled by the position on $T^{N-1}$. The statement is heuristic: the additive setting supplies no mechanism that pins $\mathrm{Re}\,\beta$, and the conjecture is retained only to mark the phase-geometric *shape* of the question, not as a target the additive apex can close.

**The concrete provable target: the multiplicative / function-field apex.** ל8 installs the layer in which an RH analog is a *theorem* rather than a conjecture. Two moves are required, and only their composition buys the theorem. First, pass from additive to **multiplicative** characters — ל8's Dirichlet characters $\psi$ of $(\mathbb{Z}/N)^\times$, with the Gauss-sum bridge $\tau(\psi, a)$ (ל8 §4) intertwining the two sectors. Second — the decisive step — change the base ring from $\mathbb{Z}/N$ (residues of $\mathbb{Q}$) to a **finite field** $\mathbb{F}_q$ (the function-field / geometric place), realized by taking multiplicative characters of $\mathbb{F}_q^\times$ of order dividing $N$, which exist precisely when $N \mid q-1$. Over $\mathbb{F}_q$ the abstract Gauss-sum bridge becomes the genuine Gauss sum $\tau_q(\chi) = \sum_{t \in \mathbb{F}_q^\times} \chi(t)\, \psi_{\mathrm{add}}(t)$ (with $\psi_{\mathrm{add}}$ a fixed nontrivial additive character of $\mathbb{F}_q$), whose modulus $|\tau_q(\chi)| = q^{1/2}$ is already a critical-modulus statement, and the Jacobi sums $J(\chi^a, \chi^b) = \sum_t \chi^a(t)\, \chi^b(1-t)$ (with $|J| = q^{1/2}$ for nontrivial factors of nontrivial product) are *the Frobenius eigenvalues* on $H^1$ of the Fermat curve $C_N : x_0^N + x_1^N + x_2^N = 0$ over $\mathbb{F}_q$ (Weil 1949). The object that carries a provable RH is then the finite $L$-function attached to multiplicative-character amplitudes encoding point-counts,

$$Z(C_N/\mathbb{F}_q, T) = \exp\Big( \sum_{n \ge 1} \frac{\#C_N(\mathbb{F}_{q^n})}{n} T^n \Big) = \frac{P(T)}{(1-T)(1-qT)}, \qquad P(T) = \prod_{i=1}^{2g} (1 - \alpha_i T),$$

$g = \frac{(N-1)(N-2)}{2}$, with $\#C_N(\mathbb{F}_{q^n}) = q^n + 1 - \sum_i \alpha_i^n$ and the inverse roots $\alpha_i$ running over the Jacobi sums. **Weil's RH for curves** (Weil 1948, 1949; Deligne 1974 in arbitrary dimension) is the theorem $|\alpha_i| = q^{1/2}$; under $T = q^{-s}$ this is exactly $\mathrm{Re}\, s = \frac{1}{2}$ — a genuine critical line, proved. The categorical multiplicative apex $\mathcal{D}_x(\beta, \psi)$ of ל8 is the $\mathbb{F}_q \rightsquigarrow (\mathbb{Z}/N)^\times$ shadow of this object; the concrete forward door is to build its $\mathbb{F}_q$ lift and read Weil's theorem off the apex (see ל8 §7.3).

**Relation to the standing simplicial-RH programme.** ל20's governed-closure architecture (with the primorial $\mathbb{Z}/6$ Hurwitz collapse and the Paper III Gram–Bures bridge) targets the *classical* RH for $\zeta$ over $\mathbb{Q}$ — the Archimedean place, where the critical line remains conjectural and must be reached by a cofinal Archimedean lift. The function-field apex is the complementary *finite / geometric* place: there the critical line needs no lift, because the $L$-function is already a polynomial whose zeros are pure of weight one — Frobenius supplies for free the purity that the Archimedean programme must manufacture. The function-field multiplicative apex is therefore the programme's worked, provable instance of its own RH-shaped door, and a sharp localization of *why* the classical target is hard: it is the missing-Frobenius place.

Forward direction (planned forward paper, number to be assigned): **The Function-Field Multiplicative Apex and Weil's Theorem — A Provable Critical Line for the Categorical Simplex.**

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## 8. ל1 through ל6 as worked cells through the ל7 lens

ל0 §7 placed ל1–ל6 as worked cells of the apex surface $\mathcal{M}_x(\beta, c)$. The complex extension promotes each cell to a slice through the larger phase-bundle structure; we re-locate the six papers within ל7's framework.

### 8.1 ל1 — Native Simplicial Chart, as the decoherent face

ל1's chart on $\mathring{\Delta}^{N-1}$ via the square-root map $\sqrt{p} : \Delta^{N-1} \to S^{N-1}_+$ identifies the simplex with the positive face of the real sphere, with Fisher–Rao the pulled-back round metric. In the ל7 framing, this is the **decoherent face** of the complex amplitude space (Definition 1.2), and the chart is the canonical "no-phase" section $\sigma_{\mathrm{R}+}$ of the Hopf-and-toric fibrations (§1.3). The factor of 2 in ל1's Fisher–Rao distance is the projective-angle doubling of Theorem 2.1.

### 8.2 ל2 — Cumulant Tower, as the moment diagonal on the decoherent face

ל2's cumulant tower at the uniform $u \in \Delta^{N-1}$ samples the moment diagonal (ל0 Theorem 3.3) at $\beta = 0, c = 0$. In the ל7 framing, this is the moment diagonal of $\mathcal{M}_z(\beta, c)$ restricted to the decoherent face: the **classical** moment tower is the phase-zero sample of the complex moment tower, with the off-equator complex extension producing additional phase-twisted moment content (Proposition 4.3's coherence content at order $n$).

### 8.3 ל3 — Character Orthogonality, as the conductor index of the complex apex

ל3's character orthogonality on $\mathbb{Z}/N$ activates the conductor index $c$. The complex extension preserves ל3's selection rule $a + b + c \equiv 0 \pmod N$ (Theorem 5.5) — the Kummer-cover deck-group action is independent of $c$, acting uniformly across all conductor modes. ל3's conductor packets $H_d$ remain the eigenspaces of the character decomposition.

### 8.4 ל4 — Helmert-Simplicial Residue Ladder, as the rational basis dual to the Kummer cover

ל4's Helmert ladder is the rational orthogonal basis of $H_N$. In ל7, the Helmert basis is the **rational descent** of the Kummer cover: each Helmert vector $e_k$ corresponds to a rational character of the toric quotient $T^{N-1}$ acting on $\mathbb{CP}^{N-1}$, and the radical clock $K_N$ records the field arithmetic of the Helmert normalizers $\sqrt{k(k+1)}$ — a rational/radical shadow of the complex/transcendental phase content of $T^{N-1}$.

### 8.5 ל5 — Harmonic-Simplicial Dual Gauge, as the parity reading of the complex apex

ל5's $\beta \to -\beta$ parity decomposition extends naturally to $\mathcal{M}_z(\beta, c)$ in the complex variable: the parity pair $\mathcal{M}_z(\beta, c) + \mathcal{M}_z(-\beta, c)$ doubles the even-$\beta$ Fourier content on $T^{N-1}$, $\mathcal{M}_z(\beta, c) - \mathcal{M}_z(-\beta, c)$ doubles the odd. On the decoherent face this recovers ל5's classical parity; off the decoherent face the parity reading is phase-twisted by the Bloch-rotation angle of Proposition 6.2.

### 8.6 ל6 — RMS-Simplicial Quadratic Gauge, as the $m = 2$ Born rung of the closure diagonal

ל6's $\beta = 2$ RMS / Born closure is the $m = 2$ closure diagonal of the complex apex. In ל7, this is the $\rho_2 = \chi_{-2 \bmod N}$ deck-group character of the Kummer cover (Theorem 5.3). For $N = 2$ (qubit), $\rho_2 = \chi_0$ is trivial — Born closure descends to the Bloch sphere without obstruction (§6.3). For odd $N \geq 3$, $\rho_2$ is non-trivial — Born closure acquires a $\mu_N$ branch obstruction in the complex extension, which the decoherent restriction suppresses. ל6 Conjecture 4 (complex amplitude extension) is the ל7 program; Theorem 4.2 + Theorem 5.3 jointly close the conjecture as a structure theorem.

### 8.7 Grid summary

| Paper | ל7 slice | Worked content |
|---|---|---|
| ל1 | decoherent face / square-root section | Fisher–Rao $= \frac{1}{4}$ Fubini–Study restriction |

| Paper | ל7 slice | Worked content |
|---|---|---|
| ל2 | moment diagonal restricted to decoherent face | classical cumulant tower as zero-phase content of complex moments |
| ל3 | conductor index $c$ of $\mathcal{M}_z$ | selection rule preserved by Kummer cover |
| ל4 | rational descent of toric $T^{N-1}$ | Helmert basis as radical shadow of phase content |
| ל5 | parity reading of $\mathcal{M}_z$ in $\beta$ | classical parity on decoherent face, phase-twisted off-face |
| ל6 | closure diagonal $m = 2$, $\rho_2$-Kummer character | Born closure = decoherent restriction of complex $m = 2$ rung |
| ל7 | the complex bundle itself | $\mathcal{M}_z$, decoherence theorem, Kummer-cover theorem, qubit anchor |

Six worked cells of the decoherent face; ל7 names the bundle they sit inside.

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## 9. Coda

The complex amplitude extension reveals the apex as a four-layer structure: a positive ledger $x \in \mathbb{R}^N_{>0}$ (ל1–ל6), its Fourier-Mellin transform on $\mathbb{Z}/N$ (ל0), the radial-and-phase-bundle structure $\mathbb{C}^N \to \mathbb{CP}^{N-1} \to \Delta^{N-1}$ (ל7), and the natural Galois/Kummer cover that becomes literal in the complex extension. The *pure-state phase redundancy* of finite-dimensional quantum mechanics on $\mathbb{C}^N$ is the cover structure, made non-trivial by the phase fiber.

The one-sentence apex of ל7:

> The pure-state phase structure of finite-dimensional quantum mechanics on $\mathbb{C}^N$ is the Hopf-and-toric covering of the categorical-simplex apex of ל0.

with global phase the Hopf $U(1)$, relative phase the toric $T^{N-1}$, and decoherence the modulus-restriction projection to the real-positive sector — the decoherent face on which $\mathcal{M}_z$ reduces to $\mathcal{M}_x$ exactly.

**Scope.** The cover is a statement about *phase redundancy*, not about all of quantum mechanics. It does not encode the non-commutativity of observables, measurement back-action, unitary dynamics, or the tensor-product/entanglement structure of multi-particle systems; those are the forward doors of §7. The two slogans below are to be read in this scoped sense.

ל0 named the surface; ל7 names the phase bundle over it.

And the opening sentence, restated (in the scoped sense above):

**Decoherence is closure of phase. Quantization is the cover.**

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